LOCAL CLIMATE CHANGE PLAN OF THE PROVINCE OF LIMA 2021-2030

CITY SERVICES AND ENVIRONMENTAL MANAGEMENT MANAGEMENT

DEPUTY MANAGEMENT OF ENVIRONMENTAL STRATEGY AND CLIMATE CHANGE

April 2021

Content

TABLE INDEX----------7
ACRONYMS ----- ----------9
GLOSSARY ----- ----------
----------13 PROLOGUE---- ----------
----------17 INTRODUCTION--
---------- ----------19 WHY A PLCC?---
---------- ----------20 SUMMARY----------
---------- ----------22 1. GENERAL
ASPECTS---------- ---------- ----------2. 3

1.1. MLEGAL ARCH..... 23
1.1.1. International regulatory framework ..... *23* 1.1.2. National regulatory framework ..... *24* 1.1.3. Local regulatory framework..... *28*
1.2. CCOMMITMENTS ASSUMED BY THE MUNICIPALITY METROPOLITANA DE LHAVE..... 29
1.2.1. On an international level ..... *29* 1.2.2. At the national level..... *30*
1.3. ALCANCE..... 31
1.4. METHODOLOGY..... 32
1.4.1. Phases of the methodological process *33*
1.4.2. Plan implementation process..... *37* 1.4.3. Plan validation ..... *37* 1.5. GCLIMATE OBERNANCY..... 38 1.5.1. Role of the Metropolitan Municipality of Lima ..... *40* 1.5.2. Actors for the comprehensive management of climate change in the province of Lima ..... *46* 1.6. CCHARACTERISTICS OF THE FIELD OF STUDY ..... 50 1.6.1. Physical and geographical ..... *50* 1.6.2. Climate ..... *53* 1.6.3. Water resources..... *56* 1.6.4. Ecosystems ..... *62* 1.6.5. Economical ..... *73* 1.6.6. Social..... *76* 1.6.6.1. Education..... *78* 1.6.6.2. Health ..... *79* 1.6.7. Cultural..... *80* 1.6.8. Mobility..... *82* 1.6.8.1. Public transportation ..... *83* 1.6.8.2. Cycle paths ..... *84* 1.6.9. Eco-efficiency ..... *85* 1.6.9.1. Electric Energy in Metropolitan Lima..... *87* 1.6.9.2. Implementation of eco-efficiency measures in the MML *89*

1.6.10. Waste 90
1.6.10.1. Current situation of solid waste management in the province of Lima..... 91

2. DIAGNOSIS ---------- ---------- 100

2.1. MITIGATION 100
2.1.1. Greenhouse gas inventory..... 100 2.1.2. (GHG) emissions scenarios..... 104
2.2. ADAPTATION: AANALYSIS OF RRISK CLIMATIC (ARC)..... 110
2.2.1. Description of climate hazards ..... 112 2.2.2. Defining climate risk..... 113 2.2.3. Hazard Analysis..... 121 2.2.4. Exposure and vulnerability analysis ..... 125 2.2.5. The climate risk 130
2.3. CAIR QUALITY 137
2.3.1. Noise pollution..... 141 2.4. INCLUSION IN THE FACE OF CLIMATE CHANGE 144
PANDEMIA OF COVID-19: CHANGES IN THE DYNAMICS OF THE CITY..... 150 2.5. BREJECTS..... 154
2.6. NECESITIES..... 156
2.7. PRIORITIES 158

3. DESIRED FUTURE ---------- ---------- 160 4. PLCC PROGRAMMING---------- ---------- ------ 167

4.1. INISSION AND OBJECTIVES OF THE PLCC AL 2030..... 167 4.1.1. Vision ..... 167 4.1.2. Goals 167
4.2. ACTIONS ANDSTRATEGIC PLCC..... 170 4.3. PANSWER ..... 209 4.4. ARTICULATION OF THE PAND LLOCATION OF CAMBIO CLIMATIC OF THE PROVINCIA DE LHAVE 2021-2030 WITH POLICIES AND PLANS 209

5. MONITORING AND EVALUATION ---------- ---------- 211

5.1. RESPONSABLE..... 212
5.2. FFOLLOW-UP AND REPORTING FREQUENCY ..... 212 5.2.1. A nivel local:..... 212 5.2.2. On an international level: ..... 213 5.2.3. At the national level: ..... 214 5.3. ANDVALUATION..... 214

6. RECOMMENDATIONS ---------- ---------- 219

7. BIBLIOGRAPHY ---------- ---------- ---------- 221

8. ANNEXES ---------- ---------- ---------- 224

INDEX OF FIGURES

FIGURE 1. METHODOLOGICAL PROCESS FOR THE FORMULATION OF THE PLCC. -------33

FIGURE 2. REPRESENTATION OF THE METROPOLITAN ENVIRONMENTAL MANAGEMENT SYSTEM (SMGA).---------- ---------- Four. Five

FIGURE 3. GEOGRAPHICAL FRAME. ---------- ----------50

FIGURE 4. DISTRICT DIVISION.---------- ----------51

FIGURE 5. GEOMORPHOLOGICAL UNITS.----- ----------52

FIGURE 6 . RELIEF AND STRUCTURE OF BASINS. ----------53

FIGURE 7. MAXIMUM ANNUAL TEMPERATURE. ---------- ---54 FIGURE 8. MINIMUM ANNUAL TEMPERATURE. ---------- ----55 FIGURE 9. TOTAL AVERAGE ANNUAL PRECIPITATION. ----------56

FIGURE 10. MONTHLY VARIATION OF THE FLOW OF THE CHILLÓN, RÍMAC AND LURÍN RIVERS.---------- ---------- ----------57

FIGURE 11. LOCATION AND LIMITS OF THE CHILLÓN RÍMAC LURÍN INTERREGIONAL BASIN WATER RESOURCES COUNCIL.---------- ----------58

FIGURE 12. WATER SOURCES FOR LIMA. ---------- --59

FIGURE 13. MAP OF ECOSYSTEMS OF THE PROVINCE OF LIMA. ----------62

FIGURE 14. STATE OF THE COASTAL HILLS OF THE PROVINCE OF LIMA. ------64

FIGURE 15. ENVIRONMENTAL MANAGEMENT UNITS OF THE ZRE PV. ----------68

FIGURE 16. MAP OF THE WETLANDS OF METROPOLITAN LIMA BASED ON A RECORD MADE BY THE NGO COOPERACCION.---------- ----------70

FIGURE 17. EVOLUTION OF THE UNEMPLOYMENT RATE, ACCORDING TO MOBILE QUARTERS 2014 – 2020. ----------74

FIGURE 18. CONCENTRATION OF ESTABLISHMENTS AND JOBS. ----------75

FIGURE 19. AGRICULTURAL VALLEYS.----------

----------76 FIGURE 20. POPULATION PYRAMID. ---------- ----------77 FIGURE 21. EVOLUTION OF THE URBAN FOOTPRINT 1954-2014.---------- ---78 FIGURE 22. DENSITY OF EDUCATIONAL CENTERS. ----------79

FIGURE 23. LOCATION OF TOURIST RESOURCES AND RURAL TOURISM ENTERPRISE. ---------- ----------81

FIGURE 24. ELECTRICAL POWER GENERATION PROCESS.----------85

FIGURE 25. PERCENTAGE OF ELECTRICAL ENERGY PRODUCTION BY SOURCES. ---------- ---------- 86

FIGURE 26. EVOLUTION OF ELECTRICAL ENERGY PRODUCTION.----------86

FIGURE 27. GENERATION OF MUNICIPAL SOLID WASTE IN THE PROVINCE OF LIMA (2010-2019)---------- ----------91

FIGURE 28. GENERATION OF MUNICIPAL SOLID WASTE BY DISTRICT – 2019. ----------92

FIGURE 29. ESTIMATED SOLID WASTE COMPOSITION IN THE PROVINCE OF LIMA.---------- ---------- ----------93

FIGURE 30. CONTROLLED MUNICIPAL SOLID WASTE IN SANITARY LANDFILLS IN THE PROVINCE OF LIMA (2010-2019).---------- ----------95

FIGURE 31. DISPOSAL OF SOLID WASTE GENERATED IN THE CITY OF LIMA.---------- ---------- ----------96

FIGURE 32. RECOVERED AND VALUED WASTE (ORGANIC AND INORGANIC) IN THE PROVINCE OF LIMA FROM 2014 TO 2018----------98

FIGURE 33. WASTE GENERATED VERSUS WASTE RECOVERED AND VALUED.---------- ---------- ----99

FIGURE 34. RECOVERED AND VALUED WASTE (ORGANIC AND INORGANIC) LIMA PROVINCE 2018.---------- ----------99

FIGURE 35. COMPARISON OF RECOVERED AND VALUED WASTE IN 2018. ---------- ---------- ---------- 100

SOURCE: (MINAM, 2019). ---------- ----------

100 FIGURE 37. GHG EMISSION SCENARIOS IN THE PROVINCE OF LIMA. -------- 105

FIGURE 38. SCENARIO OF EXISTING AND PLANNED ACTIONS. ---------- 106
FIGURE 39. SCENARIO OF AMBITIOUS ACTIONS. ---------- 107
FIGURE 40. EXTENDED SCENARIO. ---------- --------- 108
FIGURE 41. RESIDUAL EMISSIONS BY SECTOR ---------- ------ 109
FIGURE 42. FORMULA FOR DETERMINING CLIMATE RISK------- 111
FIGURE 43. CONCEPTUAL SCHEME FOR DETERMINING CLIMATE RISK.---------- ---------- -------- 111
FIGURE 44. RECORD OF EMERGENCIES AND HAZARDS BETWEEN 1970 AND 2017.----- 114 FIGURE 45. MAXIMUM ANNUAL TEMPERATURE. ---------- 115 FIGURE 46. MINIMUM ANNUAL TEMPERATURE. ---------- 116 FIGURE 47. TOTAL AVERAGE ANNUAL PRECIPITATION.---------- 117

FIGURE 48. MAXIMUM ANNUAL TEMPERATURE. CPR SCENARIO 4.5. ---------- 118 FIGURE 49. MAXIMUM ANNUAL TEMPERATURE. CPR SCENARIO 8.5. ---------- 118 FIGURE 50. MINIMUM ANNUAL TEMPERATURE. CPR SCENARIO 4.5. ---------- 119 FIGURE 51. MINIMUM ANNUAL TEMPERATURE. CPR SCENARIO 8.5. ---------- 119 FIGURE 52. ACCUMULATED ANNUAL PRECIPITATION. RCP SCENARIO 4.5 ---------- 120 FIGURE 53. ACCUMULATED ANNUAL PRECIPITATION. RCP SCENARIO 8.5.---------- 120 FIGURE 54. MAP OF HEAT ISLANDS. ---------- ------ 125
FIGURE 55. VULNERABILITY OF URBAN HABITAT TO FLOODS. ---------- ---------- ---------- 128
FIGURE 56. VIABILITY VULNERABILITY TO MASS MOVEMENTS. ---------- ---------- ---------- 129
FIGURE 57. VULNERABILITY OF URBAN HABITAT TO DROUGHT.------- 129
FIGURE 58. VULNERABILITY OF URBAN HABITAT TO HEAT WAVES. SOURCE: (IDOM, 2020) ---------- ---------- 130
FIGURE 59. PM2.5 EMISSIONS BY TYPE OF SOURCE LIMA-CALLAO (TON/YEAR - 2016). ---------- ---------- ---- 140
FIGURE 60. PM2.5 EMISSIONS BY VEHICULAR CATEGORY LIMA-CALLAO

(TON/YEAR - 2016). ---------- ---------- 141

FIGURE 61. NUMBER OF MEASUREMENT POINTS BY TYPE OF APPLICATION AREAS THAT EXCEED THE NOISE ECA IN LIMA, 2019. ---------- ----- 143

FIGURE 62. GINI INDEX BY DISTRICT. ---------- ---- 145

FIGURE 63. PER CAPITA INCOME PER APPLE. ---------- 146

FIGURE 64. CONFIRMED CASES OF COVID-19, ACCORDING TO DATE OF ONSET OF SYMPTOMS, ---------- ---------- ---------- 152

PERU 2020. ---------- ---------- 152

FIGURE 65. TREND OF CONFIRMED AND SUSPECTED COVID-19 DEATHS ---------- ---------- -------- 152

AS OF DECEMBER 5, 2020.---------- ---------- 152

TABLE INDEX

TABLE 1 POLLUTING SOURCES IN THE CHILLÓN, RÍMAC AND LURÍN BASINS. ---------- ---------- ----------60

TABLE 2 LOMAS OF THE ACR-LOMAS DE LIMA SYSTEM. SOURCE: SELF MADE.---------- ---------- --------65

TABLE 3 SECTORAL LIST OF FRAGILE ECOSYSTEMS OF THE MINAGRI LOMAS COSTERAS IN THE PROVINCE OF LIMA ---------- ----------66 TABLE 4 LOCATION OF WETLANDS OF METROPOLITAN LIMA.------ ----------69 TABLE 5 EMPLOYMENT IN ECONOMIC SECTORS BY INTERDISTRITAL AREA.-----74 TABLE 6 TRAINING CENTERS BY EDUCATIONAL LEVEL AND INTERDISTRITAL AREA.----- ---------- ----------78 TABLE 7 HEALTH FACILITIES BY CATEGORY AND INTERDISTRICT ZONE. 79 TABLE 8. TOURIST ATTRACTIONS BY TYPE AND INTERDISTRICT AREA. --81 TABLE 9. AUTOMOTIVE PARK OF THE DEPARTMENT OF LIMA, 2016. ----------83 TABLE 10. PUBLIC TRANSPORTATION FLEET REGISTERED

IN SETAME, 2018. --------83 TABLE 11. BUSES OF COMPLEMENTARY CORRIDORS, 2018.----------84 TABLE 12 FINAL CONSUMPTION OF ELECTRICAL ENERGY BY SECTOR, UNIT (GWH).----- ---------- ---------- 87 TABLE 13. NATIONAL GAP IN PUBLIC CLEANING SERVICE ----------93 TABLE 14. LOCATION OF SANITARY LANDFILLS AVAILABLE IN THE DISTRICTS OF LIMA PROVINCE.--- ---------- ---------96 TABLE 15. DISPOSAL OF MUNICIPAL SOLID WASTE IN SANITARY LANDFILLS. ---------- ----------97 TABLE 16. GAP IN THE DISPOSAL OF MUNICIPAL SOLID WASTE IN THE CERCADO DE LIMA. SOURCE: MAYOR'S RESOLUTION Nº 297 ----------98 TABLE 17. DEFINITION OF THE SCOPE OF GREENHOUSE GAS EMISSIONS. ---------- ---------- 101 TABLE 18. RESULTS OF THE 2015 GHG INVENTORY BASIC LEVEL---------- 102 TABLE 19. PERCENTAGE OF REDUCTION OF GHG EMISSIONS BY SECTOR AND BY SCENARIO ---------- ---------- 110 TABLE 20. ANALYZED METEOROLOGICAL STATIONS. ---------- 116 TABLE 21. TABLE OF AREAS SUSCEPTIBLE TO FLOODING OF THE RÍMAC, LURÍN AND CHILLÓN RIVERS. ---------- ---------- 121

TABLE 22. SUBJECTS OF THE PROVINCE OF LIMA WHO MAY POTENTIALLY BE EXPOSED TO CLIMATE HAZARDS ---------- - 126 TABLE 23. ENVIRONMENTAL QUALITY STANDARDS (ECA) FOR AIR. ---------- 140 TABLE 24. VULNERABLE GROUPS IN THE PROVINCE OF LIMA: FIRST IDENTIFICATION.---------- ---------- ---------- 149 TABLE 25. ARTICULATION MATRIX OF THE PLCC WITH POLICIES AND PLANS.---- 211 TABLE 26. PLCC MONITORING MATRIX. ---------- 218

ACRONYMS

AAA Administrative Authority of Water Cañete – Fortaleza AATE Autonomous Authority of the Electric Mass Transportation System of Lima and Callao
ACP Private Conservation Areas
ACR Regional Conservation Areas
AEI Institutional Strategic Action
ASOUT Agriculture, Forestry and Other Land Uses ALA Local Water Administration Chillón - Rímac – Lurín ANA National Water Authority
ANP Protected Natural Area

APCV Costa Verde Project Authority
APS South Pacific Anticyclone
ARC Climate Risk Analysis
ASAP Action Selection and Prioritization Tool.
ATU Urban Transportation Authority for Lima and Callao BAU Business As Usual
BRT Bus Rapid Transit, for its acronym in English. CAMET Metropolitan Environmental Commission
CAP Personnel Assignment Chart
CCAC Climate and Clean Air Coalition CDD Consecutive Dry Days Index
CDP Carbon Disclosure Project
CENAPRED National Center for Disaster Prevention
CENEPRED National Center for Estimation, Prevention and Risk Reduction CEPLAN National Center for Strategic Planning
CHL Historic Center of Lima
UNFCCC United Nations Framework Convention on Climate Change
COP Conference of the Parties CRHCI CHIRILU Chillón Interregional Basin Water Resources Council, Rímac and Lurín
CVCs Short-Lived Climate Pollutants
dB Decibel
dBA Decibel A
DGCCD General Directorate of Climate Change and Desertification DGERN General Directorate of Natural Resources Strategies DIGESA General Directorate of Environmental Health and Food Safety DRELM Regional Directorate of Education of Metropolitan Lima EMAPE Municipal Toll Administrator Company of Lima EMILIMA Municipal Real Estate Company of Lima S.A.
ENCC National Strategy for Climate Change.
ENSO El Niño-Southern Oscillation
EQT Emissions Quantification Tool
ERCC Regional Climate Change Strategy
ET Transport Companies

ETCCDI Panel of Experts on Climate Change Detection and Indices.
FEN Coastal El Niño Phenomenon
PV Photovoltaics
GCoM Global Covenant of Mayors for Climate and Energy.
GHG Greenhouse Gases
GIS Geographic Information System
GN Gas Natural
GPC Global Protocol for Community-Scale Greenhouse Gas Emission Inventories
GSCGA City Services Management and Environmental Management GTM Metropolitan Technical Groups
GTMCC Metropolitan Technical Group on Climate Change GTM-NDC Multisector Working Group for the Implementation of the NDC ICLEI Local Governments for Sustainability, for its acronym in English. IGN National Geographic Institute of Peru

IMP Metropolitan Planning Institute
INDECI National Institute of Civil Defense
INEI National Institute of Statistics and Informatics of Peru INFOCARBONO National Inventory of Greenhouse Gases INGEI National Inventory of Greenhouse Gases
INGEMMET Geological Mining and Metallurgical Institute of Peru IOARR Optimization Investment Project, Marginal Expansion, Replacement and Rehabilitation
IPCC Intergovernmental Panel on Climate Change
LED Light-emitting diode, for its acronym in English.
LM Lima Metropolitana
LMCC Framework Law on Climate Change
LMP Maximum Permissible Limits
MERESE Remuneration Mechanisms for Ecosystem Services MIDAGRI Ministry of Agrarian Development and Irrigation
MINAM Ministry of the Environment
MINCETUR Ministry of Foreign Trade and Tourism
MINEDU Ministry of Education
MINSA Ministry of Health
MML Metropolitan Municipality of Lima
MRV Measurement, Reporting and Verification
MTC Ministry of Transport and Communications
MtCO2 eq Millions of tons of carbon dioxide equivalent MVCS Ministry of Housing, Construction and Sanitation MYPE Micro and Small Businesses
NAMA National Appropriate Mitigation Actions
NAP National Adaptation Plan
NDC Nationally Determined Contributions
SES Socioeconomic Level

OECD Organization for Economic Cooperation and Development SDG Sustainable Development Goals
OEFA Environmental Assessment and Supervision Body OEI Institutional Strategic Objective
WMO World Meteorological Organization
WHO World Health Organization
NGO Non-Governmental Organizations
OPD Decentralized Public Body
OSC-J Youth Civil Society Organizations OSINERGMIN Supervisory Body for Investment in Energy and Mining PAMA Environmental Management Adequacy Plan
PATPAL FBB Board of Trustees of Parque de Las Leyendas Felipe Benavides Barreda
PAUL Urban Agriculture Platform of Lima
PDC Concerted Development Plan
PDM Metropolitan Development Plan
PDU Urban Development Plan
EAP Economically Active Population
PEI Institutional Strategic Plan
PET Working Age Population
PGG General Government Policy

PGRL Regional Government Program of Metropolitan Lima PI Incentive Program for the Improvement of Municipal Management PLANMET 2040 Metropolitan Urban Development Plan 2021-2040 PLCC Local Climate Change Plan of the Province of Lima 2021-2030 PM Particulate Material
UNDP United Nations Development Program UNEP United Nations Environment Program POI Institutional Operational Plan
PRODUCE Ministry of Production
PROHVILLA Municipal Authority of the Villa Swamps
PROLIMA Municipal Program for the Recovery of the Historic Center of Lima
PROTRANSPORTE Protransporte Metropolitan Institute of Lima
PTAP Drinking Water Treatment Plant
WWTP Wastewater Treatment Plant
PUIP Industrial Processes and Product Use
CPR Representative Concentration Trajectory
RENAMI National Registry of Mitigation Measures
ROF Regulations of Organization and Functions
RSU Urban Solid Waste
SEAC Sub-Management of Environmental Strategy and Climate Change
SEDAPAL Drinking Water and Sewage Service of Lima SENAMHI National Service of Meteorology and Hydrology of Peru SERFOR National Forestry and Wildlife Service
SERNANP National Service of Natural Areas Protected by the State SERPAR Lima Parks Service
SETAME Metropolitan Taxi Service
SGA Environmental Management Sub-Management

SIGERSOL Information System for Solid Waste Management SIGMETRO Metropolitan Information and Management System SIGRID Information System for Disaster Risk Management SINAFOR National Forestry and Wildlife Management System
SINAGERD National Disaster Risk Management System SINIA National Disaster Risk Management System Environmental Information
SISOL Metropolitan Solidarity System
SMGA Metropolitan Environmental Management System
SMIA Metropolitan Environmental Information System SNGA National Environmental Management System
SUNASS National Superintendency of Water and Sanitation Services SUTRAN Superintendency of Land Transportation of People, Cargo and Goods
UCSUR Scientific University of the South
UNESCO United Nations Educational, Scientific and Cultural Organization, for its acronym in English.
WHO World Health Organization
WRI World Resources Institute
ZRE PV Special Regulation Area Villa Swamps PATPAL Board of Trustees of Las Leyendas Park

GLOSSARY

Adaptation to climate change

Process of adjustment to the actual or projected climate and its effects through interventions (measures) aimed at moderating or avoiding potential impacts and/or taking advantage of the opportunities identified in the process (LMCC, 2018).

Climate change

Climate change attributed directly or indirectly to human activity that produces a variation in the composition of the global atmosphere and that is in addition to the natural climate variability observed over comparable time periods. (LMCC, 2018).

Adaptive capacity

Also called resilience, it is the ability of social, economic and environmental systems to cope with a dangerous event, trend or disturbance, responding or reorganizing themselves so that they maintain their structure, identity and essential functions, and at the same time preserving their capacity to adapt. learning and transformation. (LMCC Regulations, 2019).

Exhibition
It is the presence of populations, livelihoods, ecosystems, watersheds, territories, infrastructure, goods and services, among others, in areas that could be impacted by dangers associated with climate change. (LMCC Regulations, 2019).

Greenhouse Gases (GHG)
Gases that make up the atmosphere, of natural or human origin, that trap the sun's energy in the atmosphere, causing it to heat up. (LMCC, 2018)

Risk management against the effects of climate change:
Risk management in the face of the effects of climate change is a process of adopting policies, strategies and concrete actions, with the aim of reducing current damage or preventing or avoiding future damage to populations and their means and ways of life. territories, ecosystems, infrastructure, goods and services, among others, or alterations in their functioning; generated by dangers associated with climate change, and the consequent losses, for whose development is carried out in accordance with the established regulations of the National Disaster Risk Management System. (LMCC Regulations, 2019).

Impact
Effects on natural or human systems. They result from the interaction between climate threats that occur in a specific period of time and the vulnerability of an exposed system (IPCC, 2014).

Adaptation measure
They are interventions planned by state and non-state actors, which consist of: actions, practices, technologies and services necessary to reduce or avoid severe alterations, losses and damages, triggered by the dangers associated with climate change in populations, livelihoods, ecosystems, basins, territories, infrastructure, goods and services, among others; as well as to take advantage of the opportunities to climate change. (LMCC Regulations, 2019).

Mitigation measures
Mitigation measures are actions adopted by state and non-state actors, such as indigenous or native peoples, in accordance with articles 12, 13 and 17 of this Regulation, which aim to reduce GHG emissions and increase GHG removals. . Mitigation measures can contribute to the implementation of NDCs to achieve long-term low-carbon development, in accordance with the provisions of article 16 of the LMCC. (LMCC Regulations, 2019).

Climate change mitigation
Human intervention to reduce sources of greenhouse gases or improve sinks (the processes, activities or mechanisms that remove a greenhouse gas from the atmosphere), in order to limit future climate change (LMCC, 2018).

NDC
The NDCs comprise the national goals determined sovereignly by the Peruvian State to reduce national GHG emissions and increase GHG removals, adapt to the effects of climate change, and contribute to the global response to climate change. (LMCC Regulations, 2019)

Danger associated with climate change
Physical phenomenon, trend or disturbance in the environment due to gradual or extreme changes in the properties of the climate; with the probability or potential of occurring in a specific place with certain characteristics and with the capacity to cause damage or loss to a subject, severely alter its functioning. These changes in climate properties can be current and future. (LMCC, 2018)

Dangerousness
Characterization of the probability and potential incidence associated with a hazard (IPCC, 2014).

Local Climate Change Plans
The PLCC are the comprehensive climate change management instruments that guide and facilitate State action at the local level; They are aligned with the ERCC of their jurisdiction and the ENCC and contribute to the implementation of the NDC. (LMCC Regulations, 2019)

Populations in vulnerable situations
The population in a vulnerable situation is comprised of women, children and adolescents, older adults, people with disabilities, indigenous or native peoples, people deprived of liberty, migrants and those in poverty, whose economic, social and cultural conditions They reduce their capacity to adapt and mitigate climate change, so they are more exposed to the impacts and risks of climate change. (LMCC, 2019)

RCP (Representative Concentration Pathway)
Modeled scenarios to reproduce different paths of socioeconomic development at a global level and their implications on the levels of GHG emissions and concentration in the

atmosphere until the year 2100 refers, indicating its associated radiative forcing (rate of change of energy per unit surface area induced at the top of the atmosphere). The greater the radiative forcing, the greater variability in climatic conditions compared to the pre-industrial period. A zero possibility of climate change due to anthropogenic causes would imply zero radiative forcing (IPCC, 2014).

To complete the latest IPPC analysis report in 2014, these four scenarios were selected:

RCP2.6: An “optimistic” scenario, which foresees a progressive decrease in the concentration of GHGs in the atmosphere until the end of the century, with an associated radiative forcing that peaks at approximately 3 W/m2 at mid-century and then decreases.

RCP4.5 and RCP6.0: Two “intermediate” stabilization pathways in which the radiative forcing is stabilized at approximately 4.5 and 6.0 W/m2.
RCP8.5: A “pessimistic” path that considers an emissions growth rate similar to that recorded over the last decades and returns a radiative forcing that reaches more than 8.5 W/m2 by 2100.

Resilience
Capacity of social, economic and environmental systems to cope with a dangerous event, trend or disturbance, responding or reorganizing themselves in a way that maintains their structure, identity and essential functions, while preserving their capacity for adaptation, learning and transformation (LMCC, 2018).

Retroffiting
Reconditioning, rehabilitation or readaptation

Risk
Interaction of exposure and vulnerability in relation to a specific hazard (IPCC, 2014).

Food safety
Predominant situation in which people have safe access to sufficient quantities of safe and nutritious food for normal growth and development and for an active and healthy life. (LMCC 2018)

Sensitivity
Intrinsic characteristics of the exposed element that increase the probability of suffering impacts due to a climate hazard, as well as its potential direct or indirect consequences. It refers to its fragility and its value (human, economic, cultural, environmental) (IPCC, 2014).

Vulnerable Subjects
Populations, livelihoods, ecosystems, watersheds, territories, infrastructures, goods and/or services, among others, that are exposed to the impact of a danger, with low adaptive capacity to face and resist the dangers associated with climate change. (LMCC Regulations 2019)

Vulnerability
Propensity or predisposition to be negatively affected. Vulnerability comprises a variety of concepts and elements that include sensitivity or susceptibility to harm and lack of capacity to respond and adapt (LMCC, 2018).

FOREWORD

Dear neighbors of the city of Lima:

The COVID-19 pandemic is leaving us great lessons regarding the urgent need to be prepared as a society to face the main challenges facing humanity.

One of them, the one with the greatest impact, is climate change, which represents an unprecedented challenge, given that it puts the lives of people, species and ecosystems on the planet at risk.

At a global level, it is observed that the impact of climate dangers caused by the increase in temperature affects people unequally, so public policies must be oriented towards the protection of the most vulnerable, so that the response to to climate change is equitable.

In this regard, in Peru there have been important advances in the fight against climate change, such as the signing of the Paris Agreement and the promulgation of the Climate Change Framework Law (LMCC) and its regulations, which provides the necessary legal framework. so that local governments include the climate variable in their management. Cities have a decisive role in addressing climate challenges because they are home to millions of people, generate large amounts of greenhouse gases and are exposed to potential climate dangers.

In this framework, the Metropolitan Municipality of Lima joins the global effort for climate and presents to citizens the Local Climate Change Plan of the Province of Lima 2021-2030, which is aligned with the Nationally Determined Contributions of our country . The objectives of the plan are to reduce greenhouse gas emissions by 30%, increase resilience, reduce the vulnerability of the population and improve the city's adaptation capabilities to possible climate dangers.

As part of climate governance, the plan is aimed at strengthening planning, communication and education systems, with the aim of guaranteeing the active participation of citizens in climate action. At the same time, it establishes a roadmap with transformative and inclusive actions compatible with the Paris Agreement and the LMCC.

Among the main contributions of the PLCC are the Climate Risk Analysis (ARC) to 2050, which is based on national climate scenarios. With this study, adaptation actions against climate dangers (mass movements, floods, droughts and heat waves) were defined, with the purpose of increasing the resilience of the city and the adaptation capacity of sectors such as: health, food security, ecosystems, urban green and risk management.

Another contribution of the plan is the Mitigation Scenarios Study, which analyzed the possible greenhouse gas scenarios for the years 2030, 2040 and 2050, which allowed

select mitigation actions for the sectors of sustainable mobility, eco-efficiency, waste management, sustainable construction and air quality.

While the Inclusion Needs Study identified the social and economic gaps in the city that cause climate change to affect people to different degrees depending on their living

conditions. This made it possible to recognize existing inequalities and introduce the inclusion approach in climate management to reorient public policies and respond appropriately.

An additional challenge that I would like to highlight is that to materialize the PLCC, we require the participation of all actors in the city. Only by working jointly with state entities, ministries, civil society, the private sector, academia, girls, boys and young people, will it be possible to confront climate change. Climate governance recognizes the important role of children and youth, given that the measures adopted in the present will have a direct impact on their future; Therefore, climate change issues have been included in the existing participation spaces in the MML. Likewise, a special version of the PLCC has been designed for girls and boys.

In this plan you can review the proposed actions and the climate roadmap that we have for the province of Lima. With this plan I reiterate the commitment of my administration to confront climate change and for Lima to be a carbon neutral and resilient city by 2050. The time has come for bold climate action and we will work together to guarantee a resilient, inclusive, healthy and sustainable Lima. sustainable for everyone.

Jorge Muñoz Wells
Mayor of Metropolitan Lima

INTRODUCTION

The Metropolitan Municipality of Lima, through this document, delivers to the citizens the Local Climate Change Plan of the Province of Lima 2021-2030 (PLCC), in compliance with

the institutional framework established for local governments in the Framework Law on Climate Change (LMCC), which confers powers to prepare, approve, monitor and update local climate change plans.

Since our country ratified the Paris Agreement, important progress has been made to address climate change, such as the promulgation of the LMCC and its regulations, and the decisive role of cities in reducing climate change has been placed on the global agenda. of greenhouse gas (GHG) emissions.

The PLCC aims to describe how the city will adapt to climate threats and reduce its GHG emissions, focusing on the most vulnerable populations under a focus on inclusion and equity. It has been prepared by the Metropolitan Municipality of Lima, with the support of the C40 Cities Climate Leadership Group through the Climate Planning Technical Assistance Program executed between 2019 and 2020.

The scope of the plan geographically includes the 43 districts of Metropolitan Lima, including its natural areas and its rural areas, which will allow for comprehensive climate management of the city, with transversal axes in vital sectors. The temporal scope includes the time period 2021-2030 with a vision to 2050. For its design, the current state of the city has been identified and projections of climate change scenarios to 2030 have been prepared, and two projections of GHG emissions for two additional time periods, 2040 and 2050.

For its formulation, an exchange of knowledge was carried out with experts from public entities and a participatory process with key social actors in the city such as girls and boys, youth, district municipalities, social organizations and academia; which enriched the process of defining objectives and actions. The relevant authorities and social actors who must be committed to achieving the city's mitigation goals and resilience objectives have been identified, in a sustainable manner.

The Metropolitan Municipality of Lima expresses its recognition to all the people and institutions that contributed to this process. Its main objective is to generate a transformational change in the city for all citizens of Lima, with an instrument that provides a new opportunity to transform it into a sustainable, livable, inclusive, equitable and prosperous city in the face of the current and future impacts of climate change. .

WHY A PLCC?

Since 1992, Peru has been part of the United Nations Framework Convention on Climate Change (UNFCCC), an international treaty that brings together global efforts to achieve the stabilization of greenhouse gas (GHG) concentrations in the atmosphere. in order to prevent anthropogenic interference (caused by humans) that is dangerous to the climate system.

Within the framework of the Convention, the Conference of the Parties (COP) is held annually. In this event, decisions are made for the application of the Convention, analyzing the problem of climate change on a global scale and establishing goals and agreements approved by the States party to the UNFCCC, with the objective of reducing GHG emissions and stopping change. climate.

At COP 21, held in 2015, the "Paris Agreement" was adopted, signed by 195 countries. This agreement established to limit the increase in global temperature to well below 2°C, and to continue efforts to further limit the temperature increase to 1.5°C. Peru, upon ratifying the Paris Agreement in 2016, adopted the commitment to reduce GHG emissions with respect to the trend scenario for the year 2030.

This commitment was updated on December 18, 2020 with the Nationally Determined Contributions (NDC) Report for the period 2021-2030. In this report, the Peruvian State commits that its net greenhouse gas emissions do not exceed 208.8 $MtCO_{2eq}$ in the year 2030 (unconditional goal) and that greenhouse gas emissions could reach a maximum level of 179.0 $MtCO_{2eq}$ depending on the availability of international external financing and the existence of favorable conditions (conditional goal).[1]

In this context, local governments like Lima acquire an important role by acting as managers of city services, as well as urban planning; given that, currently, cities host the majority of the population and are the sources of a significant percentage of the world's greenhouse gas emissions.

The proper administration of cities requires profound transformations through urban planning and the implementation of actions to reduce emissions from the transportation sector, improve energy efficiency, reduce and valorize solid waste, and develop resilience against climate change. From the identification of this need, instruments such as Local Climate Change Plans emerge; strategic documents that demonstrate how a city will deliver on its commitment to addressing climate change.

The Metropolitan Municipality of Lima is part of the cities that participate in the C40 Technical Assistance Program for Climate Planning, whose implementation was ratified through a commitment signed in February 2019 with the Mayor of Lima. In addition, it is part of the "Global Covenant of Mayors for Climate and Energy in Latin America and the Caribbean", the largest global alliance of mayors and leaders. local

[1] Contributions determined at the National Level of Peru. Update report for the period 2021 – 2030.

committed to climate action[2]. This network sees local governments as agents of change, committed and articulated with other local social actors, to carry out ambitious climate and

energy initiatives that lead cities and their inhabitants towards an inclusive, fair and resilient future, and that allows achieve the goals proposed by the Paris Agreement.

The commitments assumed as part of this global network of mayors are aligned with climate planning, as they consider the development of instruments and tools to manage climate change. In this way, the city embarked on the process of preparing the Local Climate Change Plan of the Province of Lima 2021 - 2030, in order to establish a clear and measurable roadmap to address mitigation and adaptation to climate change. that allows the city to achieve carbon neutrality by 2050.

[2] The mayor of the city of Lima, Mr. Jorge Muñoz Wells, was chosen in 2019 to represent Latin America and the Caribbean in the C40 Cities Climate Leadership Group before the Board of Directors of the Global Covenant of Mayors for Climate and Energy (GCoM).

SUMMARY

The PLCC gathers information from the city from a climate approach, which is oriented to the analysis for the reduction of GHG emissions, the development of resilience, all under a gender, intergenerational and intercultural approach. The contents allow us to understand the formulation of the Plan in accordance with the characteristics of the province of Lima.
and the current state of the city, as a starting point to address climate change management.

In the General Aspects section, the existing legal framework in Peru is presented, as well as the international commitments, the scope of the plan at the geographic level of the city, the methodology used to construct the plan from a participatory approach, the state of the governance of the city and the characteristics of the city at a physical, climatic, environmental, social, cultural level, among others.

In the Diagnostic section, the Lima (GHG) Inventory is presented with a base year of 2015, a study of GHG emissions projections, the state of the city's air quality, an analysis of possible climate risks in the city, the inclusion approach to climate change that has been integrated into the plan, and a socio-environmental analysis of the comprehensive management of climate change through the subchapters of Gaps, Needs and Priorities for the city.

In the Desired Future section, the vision that is expected to be achieved by 2050 is presented, identified as a result of the participatory process carried out for the construction of the PLCC with girls, boys and youth organizations, who will be the generations that will have to face the challenges. results of the actions adopted today to reduce emissions and increase the response, adaptation and resilience capacity of the population and the city. Mitigation, adaptation and governance objectives are also described, as well as long-term strategies to achieve what is proposed in the Plan.

In the Programming section of the Plan, the vision of the city for 2030, the mitigation, adaptation and governance objectives, are described, in which the existing actions are framed to ensure that Lima is a low-carbon city and resilient to climate change. climatic dangers. The actions are grouped by sectors of the city such as urban green, transportation, water resources, solid waste, among others, indicating the actors and vulnerable subjects of the city linked to its management.

Finally, in the Monitoring and Evaluation section, the mechanisms for the implementation of the PLCC are presented, which follows the planning structure of the MML, in accordance with national regulations, and which has been specifically adapted for this plan according to with climate management needs.

1. GENERAL ASPECTS

1.1. Legal framework

1.1.1. International regulatory framework

The United Nations Framework Convention on Climate Change (UNFCCC), adopted in New York in 1992, brings together 197 States parties, including Peru, which approved it in 1993, through Legislative Resolution No. 261853. Said convention was established with the goal of achieving stabilization of greenhouse gas (GHG) concentrations in the atmosphere at a level that prevents dangerous human-caused interference with the climate system. In this regard, Peru, as a signatory country, committed to comply with the decisions derived from the Conference of the Parties.[4](COP, for its acronym in English).

As a result, the States Parties gathered at the Conferences of the Parties (COP) year after year since 1995. At these meetings, two important international instruments for the global community were adopted:

The Kyoto Protocol[5] adopted in 1997, approved by Peru through Legislative Resolution No. 27824 and ratified by Supreme Decree No. 080-2002-RE in 2002. This instrument incorporated, by virtue of the principle of common but differentiated responsibilities, binding objectives to reduce seven GHG emissions[6] causing global warming, in particular for industrialized countries members of the Organization for Economic Cooperation and Development (OECD) of 1992, and for countries in the process of transition to a market economy (Russian Federation, the Baltic States and several Central and Eastern European States). While, for developing countries, such as Peru, the obligation to inform and report their emissions was established.

[3] As a signatory country (or Party) to the UNFCCC, like the other 194 member countries, Peru is committed to promoting the sustainable management and conservation of all GHG sinks and reservoirs; to formulate, apply, publish and regularly update national and/or regional programs that contain measures aimed at mitigating climate change; and to cooperate in preparing for the adaptation, among other contents in Article 4 of the official text. Likewise, it is subject to the decisions of the Conference of the Parties (COP) that governs said Convention as long as, and insofar as, said decisions do not contravene national laws or require additional authorizations in such cases.

[4]Conference of the Parties. It is the supreme body of the UNFCCC, where representatives of member countries meet annually to make decisions regarding the issues of the Convention.

[5] The Kyoto Protocol came into force for Peru on February 16, 2005, and its objective was to: Reduce greenhouse gas emissions from industrialized countries to a level lower by no less than 5% than in 1990, for a period of time. commitment period between 2008 and 2012. Through the Doha Amendment to the Kyoto Protocol of 2012, ratified by Peru through Supreme Decree No. 039-2014-RE, a second period is established from the year 2013 to 2020.

[6] The gases are carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and the other three are types of gases fluorinated industrial products: hydrofluorocarbons (HFC), perfluorocarbons (PFC) and sulfur hexafluoride ($SF)._6$). The Doha Amendment to the Kyoto Protocol includes a seventh gas: nitrogen trichloride (NF_3).

The Paris Agreement[7] adopted in 2015 and ratified by Peru in 2016 through Supreme Decree No. 058-2016-RE, constitutes a universal agreement on methods to reduce climate change, setting as objectives: a) maintaining the increase in temperature global average well below 2°C above pre-industrial levels, and continue efforts to limit this temperature increase to 1.5°C above pre-industrial levels; b) increase adaptive capacity to the adverse effects of climate change and foster climate resilience and low-GHG emissions development in a way that does not threaten food production; and c) ensure that financial flows are consistent with a trajectory that leads to low-emissions and climate-resilient development. In this regard, Peru, upon ratifying said Agreement, adopted the commitment to reduce 20% of its emissions by 2030, plus an additional 10% that is conditional on international cooperation. To fulfill this commitment, the State must implement mitigation and adaptation actions to climate change framed in the Nationally Determined Contributions (NDC).

The 2030 Agenda for Sustainable Development (Agenda 2030)[8] approved in September 2015 by the States party to the United Nations Organization, including Peru, and which seeks to establish a transformative vision towards economic, social and environmental sustainability under a 15-year horizon. This Agenda has 17 interrelated Sustainable Development Goals (SDGs) and 169 goals. SDG 13 “Climate Action” specifically develops goals focused on strengthening resilience and the capacity to adapt to risks related to climate and natural disasters; incorporate climate change measures into national policies, strategies and plans; as well as improving education, awareness and human and institutional capacity in relation to climate change mitigation and adaptation, the reduction of its effects and early warning.

1.1.2. National regulatory framework

The Political Constitution of Peru recognizes in paragraph 22 of article 2, the fundamental right of every person to enjoy a balanced and adequate environment for their development. Likewise, it establishes in article 55 that the treaties concluded by the State and in force are part of national law. In this sense, considering the commitments assumed by the Peruvian State through the UNFCCC, and considering the ratification of the Kyoto Protocol and the Paris Agreement, various instruments were issued to contribute to the comprehensive management of climate change at the national level, such as as:

The General Government Policy (PGG) approved by Supreme Decree No. 056-2018-PCM, according to which all plans of ministries, regional and local governments contribute to compliance with the PGG. The PLCC is aligned with Axis 3 of the Policy, Equitable, Competitive and Sustainable Economic Growth, in relation to section 3.4. which highlights promoting competitiveness based on development potential

[7] The Agreement entered into force for Peru on November 4, 2016.

[8] The 2030 Agenda was approved by the General Assembly of the United Nations, in the Agenda entitled “Transforming our world: the 2030 Agenda for Sustainable Development", which came into force on January 1, 2016.

economic of each territory, facilitating its articulation to the national and international market, ensuring the sustainable use of natural resources and heritage cultural.

The National Environmental Policy[9] approved by Supreme Decree No. 012-2009- MINAM, which through Policy Axis 1, Conservation and Sustainable Use of Natural Resources and Biological Diversity, establishes objective 7: Achieve the adaptation of the population to change climate and establish mitigation measures, aimed at sustainable development.

The Action Plan for adaptation and mitigation against climate change, approved by Ministerial Resolution No. 238-2010-MINAM, covers the roles of all actors who carry out activities linked to the generation of GHG emissions, the operation of carbon, to the study and economic and social research of the risks and impacts of climate change, in addition to sectoral and regional development projects and actions that must provide for adaptation to climate change.

Provisions for the preparation of the National Inventory of Greenhouse Gases (INFOCARBONO), approved by Supreme Decree No. 013-2014-MINAM, constitute the regulatory and institutional framework for the collection, evaluation and systematization of information related to the emission and removal of greenhouse gases. GHG. The National Greenhouse Gas Inventory (INGEI) is made up of information from the Annual GHG Reports prepared by sectoral authorities such as the Ministry of the Environment, Agriculture and Irrigation, Housing, Construction and Sanitation, Energy and Mines, Transportation and Communications, and Production.

The National Strategy on Climate Change[10], approved by Supreme Decree No. 011-2015-MINAM, reflects the commitment of the Peruvian State to act against climate change in an integrated, transversal and multisectoral manner, complying with the international commitments assumed by Peru before the United Nations Framework Convention United Nations on Climate Change (UNFCCC), and taking into special account the efforts of foresight and action to adapt productive systems, social services and the population, to the effects of climate change. Likewise, its strategic objectives are for the population, economic agents and the State to increase awareness and adaptive capacity for action against the adverse effects and opportunities of climate change; and conserve carbon reserves and reduce GHG emissions.

Guidelines for the Integrated Management of Climate Change and the Climate Management Initiative, approved through Ministerial Resolution No. 090-2016-MINAM, with the aim of contributing to transversal and integrated management against change in the context of development

[9]The National Environmental Policy is currently in the process of being updated in accordance with Ministerial Resolution No. 062-2020-MINAM "The process of updating the National Environmental Policy is declared of priority interest."

[10] In accordance with Ministerial Resolution No. 242-2019-MINAM, this instrument is part of the sectoral list of National Policies under the guidance or leadership of the Ministry of the Environment.

sustainable development of the country at the national and subnational level, within the framework of the international commitments assumed.

The Action Plan on Gender and Climate Change of Peru, approved by Supreme Decree No. 012-2016-MINAM, is a management instrument that aims to develop and implement public policies that consider the differentiated impacts of climate change in function to gender.

Through Supreme Resolution No. 005-2016-MINAM, the Multisectoral Working Group (GTM-NDC) was created in charge of generating technical information to guide the implementation of the NDC, which was made up of the National Center for Strategic Planning (CEPLAN).) and the Ministries of Environment, Energy and Mines, Transportation and Communications, Production, Housing, among others. The GTM-NDC prepared and approved its Final Report in 2018, in which 91 adaptation measures and 62 GHG mitigation measures were defined.

The Vision of Peru for 2050 approved in session 126 by the members of the National Agreement Forum, in April 2019, determines that Peru by the year 2050 sustainably manages the territory and its ecosystem services, as well as having considerably mitigated the effects of climate change, reducing forest deforestation, implementing reforestation strategies and reducing greenhouse gas emissions. Having also implemented adaptation mechanisms to climate change, generating resilience to its effects.

Through Emergency Decree No. 029-2019, incentives were established to promote scrap metal, as a mechanism aimed at the renewal or definitive removal of vehicles from the vehicle fleet, in order to reduce GHG emissions and local pollutants that affect the public health, as well as contributing to reducing accidents on public roads and safeguarding road safety. Since, currently there is a vehicle fleet with an average age of 15.5 years for the private fleet, and 22.5 years for the public passenger and cargo transportation service fleet, which causes severe damages, such as the generation of traffic accidents, as well as vehicle traffic jams due to breakdowns on public roads, and a high percentage of emissions of polluting gases and particulate matter due to its age.

Law No. 30754, Framework Law on Climate Change (LMCC), enacted and published in 2018, establishes the principles, approaches and general provisions to coordinate, articulate, design, execute, report, monitor, evaluate and disseminate public policies. for the comprehensive, participatory and transparent management of adaptation and mitigation measures to climate change; in order to reduce the country's vulnerability to climate change, take advantage of the opportunities of low-carbon growth and comply with the international commitments assumed by the State before the United Nations Framework Convention on Climate Change, with an intergenerational approach.

Through Supreme Decree No. 013-2019-MINAM, the LMCC Regulations were approved, which defines the functions of the competent authorities in matters of climate change at the three levels of government, for planning, articulation, execution, monitoring , evaluation, reporting and dissemination of public policies for the comprehensive management of climate change. Likewise, said Regulation specifies the scope of the management instruments

comprehensive approach to climate change, explains what adaptation and mitigation measures consist of and how they are monitored, and breaks down the operation of tools for measuring, reporting and verifying GHG emissions.

Article 14 of the LMCC, in accordance with articles 18 and 20 of the LMCC Regulations approved by Supreme Decree No. 013-2019-MINAM, establishes the NDCs as an instrument for comprehensive climate change management, which include the goals national measures determined by the Peruvian State to reduce emissions and increase GHG removals, adapt to the effects of climate change, and contribute to the global response to climate change. Likewise, article 8 of the LMCC Regulations indicates that the NDCs are defined by the competent authorities in matters of climate change and are incorporated into the institutional strategic planning of the regional and municipal sectors and governments. For this purpose, the Ministry of the Environment will disseminate, through a ministerial resolution, every five years, as appropriate, the adaptation and mitigation measures to climate change, in addition to being responsible for their monitoring and evaluation; and to report on its implementation to the UNFCCC Secretariat.

Likewise, article 11 of the LMCC Regulations establishes the functions that provincial and district municipalities have, within the framework of their powers and in accordance with their capabilities, based on the principles and approaches established in articles 2 and 3 of the LMCC, one of them being, to prepare, approve, implement, monitor, evaluate and update its Local Climate Change Plan (PLCC), constituting this, in accordance with the provisions of article 18 of the LMCC Regulations, a comprehensive climate change management instrument that guides and facilitates State action at the local level, contributing to the implementation of the NDCs.

In accordance with this, article II of the Preliminary Title of Law No. 27972, Organic Law of Municipalities, provides that local governments enjoy political, economic and administrative autonomy in matters within their jurisdiction. Likewise, paragraphs 3.1 and 3.2 of paragraph 3 of article 73 establish that, in matters of environmental protection and conservation, local governments are competent to formulate, approve, execute and monitor local plans and policies on environmental matters and against the climate change, in accordance with regional, sectoral and national policies, standards and plans.

Additionally, according to the provisions of article 23 of the LMCC Regulations, the PLCC are formulated and updated by the provincial municipalities according to their capacities and progressively, in coordination with the district governments and population centers, in accordance with the guidelines prepared by the Ministry of the Environment and the current regulations on the matter. In this regard, the adaptation and mitigation measures defined in the PLCC must be included in their Local Concerted Development Plans, Institutional Strategic Plans, Institutional Operational Plans and in their budget programs. For the preparation and implementation of the PLCC, municipalities must guarantee spaces for effective, full, timely and continuous citizen participation, with state and non-state actors, such as indigenous or native peoples.

Likewise, Law No. 29973, General Law of Persons with Disabilities, has the purpose of establishing the legal framework for the promotion, protection and realization, in conditions of

equality, of the rights of people with disabilities, promoting their development and full and effective inclusion in political, economic, social, cultural and technological life.

In addition, Law No. 29535, Law that grants official recognition to Peruvian Sign Language, enacted and published in 2010, has the objective of granting official recognition and regulating Peruvian Sign Language as a language of people with disabilities. hearing throughout the national territory.

Likewise, section 3.11 of article 3 of the LMCC establishes that the climate risk approach is incorporated in the formulation of investment projects, as well as the variable of disaster risks, resilience and vulnerability to climate change in territorial planning instruments. of the regions, in order to have preventive and planned management in the face of the impacts and risks of climate change.

Additionally, article 24 of the LMCC Regulations establishes that the National Adaptation Plan, created by decision 5/CP.17 of COP17, is an instrument that contributes to the comprehensive management of climate change through implementation guidance. of adaptation measures to reduce the risks of the adverse effects of climate change and take advantage of its opportunities at the national, regional and local levels.

It should also be mentioned that on December 18, 2020, MINAM updated the Report on Nationally Determined Contributions for the period 2021-2030.

1.1.3. Local regulatory framework

Through Municipal Ordinance No. 1628, the Metropolitan Environmental Policy was approved, which determines as policy axis 2: Mitigation and adaptation to climate change, establishing as objectives; 1) Protect human life and minimize the impacts of climate change on urban infrastructure and economic activities
productive of the city; 2) Promote an eco-efficient city, in its services and in its production patterns; and 3) Reduce the ecological footprint of the city, promoting a culture of saving energy, water and materials, in public services and in private activities.

Likewise, through Municipal Ordinance No. 2208-MML, which approves the Regulation of Organization and Functions (ROF) of the Metropolitan Municipality of Lima, which empowers the City Services and Environmental Management Management, to develop the Local Climate Change Plan of the Province of Lima 2021-2030.

Through Municipal Ordinance No. 2256-MML that updates the Metropolitan Environmental Management System, comprehensive management of climate change is established as a thematic area, through adaptation and mitigation measures that are included in the Metropolitan Development Plan. (PDM), Local and Regional Development Plans, as well as in the Institutional Strategic Plan (PEI) and the Operational Plan (POI) of the Metropolitan Municipality of Lima (MML).

Through Mayor's Resolution No. 503 issued in 2019, the 2020-2023 Institutional Strategic Plan of the Metropolitan Municipality of Lima was approved, which

establishes as Institutional Strategic Action (AEI 03.05): Mitigation and adaptation and inclusion actions against effective climate change in the province of Lima, considered within the Institutional Strategic Objective (OEI 03): Implement sustainable environmental management and the conservation of ecosystems in the province of Lima.

Through Ordinance No. 2194 issued in 2019, the Master Plan of the Historic Center of Lima is approved with a vision to 2035, which contains chapters and annexes on Landscape Treatment, Sustainable Urban Mobility and Housing Recovery, among other pertinent topics. .

Through the Single Administration Regulation of the Historic Center of Lima (RUACHL) approved by Ordinance No. 2195-2019, article 5 establishes that the rules established in this regulation prevail over any other municipal rule related to the scope of the historic center. of Lima that could generate actions or intervention projects contrary to its adequate conservation; For this reason, through this Ordinance, it is established that it is mandatory for all Managements that make up the organization of the MML to have the favorable and binding opinion of the Municipal Program for the Recovery of the Historic Center of Lima, from now on. From now on it will be called - PROLIMA - before carrying out: inspections, development processes and updating of the plan, identification of priorities, changes and/or authorizations for use, development of any type of economic, cultural, traditional, tourist and social projection activity, as well as any other measure that these Managements consider appropriate for the safeguarding of cultural heritage; within the space included in the Historic Center of Lima; unless they are procedures whose characteristics and duration are regulated by law.

Through Ordinance No. 2273, which promotes universal accessibility and encourages the inclusion of people with disabilities in Metropolitan Lima, the conditions are established that allow the gradual adaptation of accessibility in the urban infrastructure of the city, so that the spaces , buildings and commercial establishments can be accessible and used by all people, regardless of their condition, exceeding the minimum standards established in national provisions, in order to promote the creation and implementation of measures that promote good safety practices. accessibility that fosters inclusion.

1.2. Commitments assumed by the Metropolitan Municipality of Lima 1.2.1.

On an international level

The Global Covenant of Mayors for Climate and Energy is the largest global alliance of mayors and local leaders committed to climate action, with the aim of accelerating ambitious and measurable climate and energy initiatives that lead to an inclusive, fair, sustainable future. low emissions and climate resilient, which helps meet and exceed the objectives of the Paris Agreement.

Regarding this purpose, the Metropolitan Municipality of Lima, through the letter of intent dated May 20, 2019, assumed the commitment to develop and report a GHG inventory at the municipal scale and an evaluation of climate risks and vulnerabilities;

plans to address climate change mitigation and adaptation, sustainable energy access, including provisions for regular assessments and reviews; among others. This commitment is reinforced by the role adopted by the current Metropolitan Mayor of Lima, who assumed the representation of the cities of Latin America and the Caribbean and the Climate Leadership Group of C40 Cities before the Board of Directors of the Global Compact of Mayors for Climate and Energy (GCoM) for 2 years, starting in September 2019.

Through Council Agreement No. 375 of 2015, Memorandum of Understanding No. 1546-2015-MML-GMM was approved between the MML and the C40 Climate Leadership Group, made up of a group of cities from around the world. that join efforts and share experiences to reduce GHG emissions and to implement adaptation mechanisms to climate change. With the Development Document of the Agreement of the Metropolitan Municipality of Lima and C40 signed in February 2019, the MML reaffirmed its commitment to chart the path towards being a carbon neutral and climate resilient city by the year 2050, with technical assistance of C40 and the involvement of citizens, business leaders and institutions. Furthermore, in order to continue counting on the support of C40 Cities, in December 2020 the MML renewed the Cooperation Agreement between both institutions. Likewise, as part of the C40 network, the MML is part of the Good Food Declaration, which seeks to develop sustainable, inclusive and resilient food systems, including the measures and strategies for their achievement in the Lima Climate Action Plan; In addition, the MML is part of the Clean Air Cities Declaration, which aims to understand the problem, determine ways to control air pollution, protect people and evaluate the impacts on health.

The Climate and Clean Air Coalition (CCAC) aims to substantially reduce or avoid methane and carbon emissions through better waste management practices. In such a scenario, the MML, through the Letter of Intent of November 2017, has committed to dedicating resources and sharing information to ensure the implementation of concrete and prioritized actions to reduce emissions from the waste sector. As well as collaborate with CCAC on the city's urban solid waste (MSW) program by collecting and sharing relevant data to inform mitigation actions and using the Emissions Quantification Tool (EQT) for evaluation and documentation of reductions in Living Climate Pollutants Short (CVCs) expected. In addition, the MML is part of the global “BreatheLife” campaign, an initiative led by CCAC, the World Health Organization (WHO) and the United Nations Environment Program (UNEP), which seeks address the air pollution crisis and work towards safer air for all.

1.2.2. At the national level

The “Pact for Public Spaces”, managed by the citizen observatory Lima Como Vamos and Ocupa Tu Calle, aims to improve urban public policy in relation to the management and quality of public spaces, providing guidelines for action, commitments and goals. The MML

assumed 11 of the 15 commitments part of the pact in December 2019, which includes, among other aspects, the sustainable mobility work axes, prioritizing pedestrians, cyclists, micromobility and public transport over automobiles; and a commitment related to the Environment and the City, for the protection of natural-urban ecosystems that ensures the resilience of vulnerable populations to the impacts of climate change.

The commitment document “For Sustainable Environmental Management in Metropolitan Lima”, signed by the MML and other districts of LM in the month of March 2019, aims to articulate the district municipalities of Lima for the exercise in an adequate and timely performance of their functions to ensure the right to enjoy a balanced environment. The MML is one of the 28 municipalities part of the commitment, which contemplates actions for the incorporation of mitigation and adaptation measures to climate change in the Concerted Development Plan (PDC), Urban Development Plan (PDU), Institutional Strategic Plan (PEI) and Institutional Operational Plan (POI), this in accordance with the ENCC, the NDC and the territorial management instruments, in accordance with the LMCC Regulations.

1.3. Range

The Scope of the Local Climate Change Plan of the Province of Lima includes geographical and temporal aspects. The first of them refers to the territorial district of the province of Lima, including its 43 districts, which will allow for comprehensive climate management of the city, with transversal axes in vital aspects such as water resources, ecosystems, energy. , urban mobility, food security, health, among others.

The province of Lima has as one of its differentiating characteristics that, at a political level, it is made up of 43 district municipalities, as previously noted, which have autonomy to exercise government, administrative and administrative acts, and have been considered in the Plan, under the same climate vision, including, for example, a Climate Risk Analysis under a spatial and geographical approach. As part of the design of the PLCC, the district municipalities were included in the participatory process for the formulation of the Plan, allowing them, in turn, to play their role in the management of climate change in the city.

On the other hand, regarding the temporal aspects, the scope of the Plan corresponds to the period of years 2021-2030, and it is based on this time horizon that a vision, strategic objectives and actions oriented to the emissions reduction goals have been designed. and adaptation. It is also aligned, on a temporal level, with the Nationally Determined Contributions (NDC) presented by the Ministry of the Environment (MINAM) to the United Nations Framework Convention on Climate Change (UNFCCC). .

To understand the roadmap that the city of Lima must follow to become a carbon neutral and resilient metropolis, a series of emissions projection scenarios have been developed as part of the mitigation baseline. The scenarios contemplate analyzing the reduction of emissions in the short, medium and long term, for the years 2030, 2040 and 2050, respectively.

Likewise, a climate risk analysis for 2050 was carried out based on national climate scenarios. The 2050 vision has the benefit of identifying ambitious actions, understanding

the opportunities and barriers on the path to carbon neutrality and climate resilience of the city, laying the foundation for managing climate change with a longer time vision and aligning the PLCC with the new urban planning instruments that the city is developing, such as the Metropolitan Plan of

Urban Development 2021-2040 (PLANMET 2040) and the Concerted Development Plan of the Province of Lima 2021-2035, which are currently being formulated by the MML.

The need to work with a long-term vision lies in the fact that the effects of climate change are cumulative over time, therefore, future generations will be affected by the decisions and policies adopted in the present, in view of which it is vital to implement actions from an intergenerational approach. Likewise, unlike other scientific disciplines, the climate change variable requires the generation of information over long periods of time, for its inclusion in studies, climate risk analysis, projection of emissions scenarios, GHG inventories, among others. main.

Finally, as part of the PLCC, a vision towards the year 2050 is included, so that this instrument is aligned with global trends and alerts, from the UNFCCC and the international scientific community, which point out the need to define ambitious goals for reducing emissions. and the development of capacities to face potential climate dangers for the middle of this century. These goals must be defined and led by the national, regional and local governments of the member countries. For this reason, the MML aligns itself with the international agreements adopted by Peru and assumes its role, within the scope of its powers, in providing citizens with the mechanisms that allow and guarantee, in the long term, that the city is habitable in a safe, inclusive and healthy way for the population.

1.4. Methodology

The methodology followed for the formulation of the Local Climate Change Plan of the Province of Lima 2021 - 2030 is framed in the Framework Law on Climate Change (LMCC)[11] and its regulations[12], the guidelines of the C40 Climate Action Planning System[13], the national and local planning regulations in force in Peru, as well as the context and needs of the city.

The Metropolitan Municipality of Lima had the support of the C40 Technical Assistance Program for Climate Action Planning, which made available a wide range of activities for the plan formulation process, such as training, academies for collaboration between local and regional peers, workshops, consultancies for the development of the technical studies required by the PLCC and its climate planning tools; with the objective that the plan is designed in accordance with the Paris Agreement, Peruvian regulations and the participation standards of the technical assistance program.

In the C40 Climate Action Planning System, a climate action plan is defined as a document, or a series of documents, that demonstrates how the city will deliver, within the framework of the Paris Agreement, its commitment to address climate change, which must

include the following parameters:

[11] Law No. 30754
[12] Approved by Supreme Decree No. 013-2019-MINAM
[13] Available in Link

- Develop a path to carbon neutrality by 2050 at the latest for the city, and set intermediate targets;
- Demonstrate how the city will adapt and improve its resilience to climate risks that affect the city currently and will do so in the future; • Inform the community, presenting the social, environmental and economic benefits expected with the implementation of the plan, and establishes ways to ensure an equitable distribution of benefits;
- Explain the governance and competencies in the city, its capacity, in addition to identifying the partners who must participate in order to accelerate the fulfillment of the mitigation goals and resilience objectives in the city.

1.4.1. Phases of the methodological process

In accordance with what was previously described, the methodology for the formulation and implementation of the plan has followed the guidelines provided in said system, which consists of the phases presented in Figure 1.

Figure 1. Methodological process for the formulation of the PLCC.
Source: (C40, 2018).

1.4.1.1. Start of the process

The first phase began with the MML's commitment to chart the path towards carbon neutrality and the city's resilience by 2050 with the technical assistance of C40 and the involvement of citizens. This was formalized through the signing of the Agreement Development Document, which details the development of the Agreement between the Metropolitan Municipality of Lima and C40 in February 2019.

The GSCGA, through the SEACC, in response to its functions, to formulate climate change management plans, was in charge of developing the PLCC; in turn, it acted as counterpart to the Climate Planning Technical Assistance Program. In this

context, the SEACC received the Advisor for the City, in charge of providing the necessary support for the development of the Plan.

Then two internal groups of the Municipality were convened between February and March 2019; a technical group in charge of working and participating in the generation of information and climate actions, which includes experts on various topics, and a governance group made up of officials in the positions of managers and assistant managers for making decisions related to planning. climate of the city.

Finally, the Strategic Evaluation of Climate Planning of Metropolitan Lima was carried out, which analyzed the city's progress in managing climate change. As part of this study, the available climate information, the existing climate information gaps and the competencies of the MML, and other relevant social actors in the province of Lima, in relation to climate change, were identified. During this phase, more than a dozen interviews and two workshops were carried out in April 2019 with actors internal and external to the municipality, with the purpose of refining the information obtained from secondary sources, as well as to communicate to the city, the beginning of the PLCC formulation process.

1.4.1.2. Preparation of the baseline

The second phase consisted of preparing the baseline, which collects information on three topics: mitigation, adaptation and inclusion. Regarding climate change mitigation, the MML had the GHG Inventory of the province of Lima with a base year of 2015, which was used as input to build three projection scenarios for GHG emissions to 2050 using the Pathways tool.[14]. As part of this process, a workshop was held, as well as technical meetings with actors internal and external to the municipality in July 2019 with the objective of validating the information and assumptions included in said tool. The GHG emissions projection scenarios made it possible to analyze the city's ambition and establish mitigation strategies, actions, goals and objectives.

For the adaptation baseline, the Climate Risk Analysis (ARC) was carried out for the province of Lima, which analyzes four climate hazards: mass movements, floods, drought and heat waves. The risk is calculated by future changes in hazards based on the climate scenarios developed by SENAMHI for the Third National Communication on Climate Change (MINAM, 2016) for RCP 4.5 and 8.5.[15]; which is then spatially crossed with the current exposure and vulnerability of the city. The methodology and results were reviewed by SENAMHI, MINAM and MML experts through three virtual meetings during the first months of 2020.

[14] Pathways is a tool developed by C40 that aims to help cities define necessary strategies to reduce GHG emissions. The tool allows you to analyze and define information and assumptions for the construction of different future scenarios for the city.

[15] RCP refers to Representative Concentration Paths (RCP), which describe the total Radiative Forcing (RF) for the year 2100 expected based on different GHG emission levels. The IPCC in its Fifth Report establishes RPC 4.5 as a scenario of stabilization of emissions and RCP 8.5 as a scenario where emissions continue to grow.

Finally, for the inclusion baseline, the Inclusion Needs Study for Climate Action was carried out, which identifies the groups and communities vulnerable to climate change in Metropolitan Lima, analyzes the strengths and weaknesses of the city around the issue. and identifies mechanisms for the inclusion of vulnerable actors in the climate action planning and implementation process. Finally, it provides a series of inclusion indicators for monitoring and evaluating the Plan in the short and medium term.

1.4.1.3. Identification of objectives, goals and actions

The third phase consisted of identifying the strategies, actions, goals, objectives and vision for the plan based on the gaps and needs that were identified in the previous stage. With all the information collected, a long list of PLCC actions was created. This list includes existing actions, which are planned by the MML, as well as new actions incorporated from the main results obtained from the baseline development process.

The actions are categorized into three groups:

- **Transversal actions:** oriented to climate governance, communication and education.
- **Mitigation actions:** aimed at reducing emissions, includes actions in the energy and

industrial processes, waste, mobility and air quality sectors.

- **Adaptation actions:** aimed at adaptation and reducing vulnerability to climate change, includes actions in the water and water resources sectors, risk management, health and food security, ecosystems and green areas, infrastructure and vital services.

In the case of adaptation actions, these were reviewed in the same baseline workshop in July 2019. Then, in June 2020, a series of virtual adaptation workshops were held aimed at internal and external actors of the MML, in which the results of the Climate Risk Analysis, the existing actions and the new actions identified and proposed for the Plan were presented,

The adaptation workshops were organized thematically, which allowed the actions to be reviewed through their grouping into five themes: water and wastewater, ecosystems and urban greenery, health and food security, infrastructure and vital services, and risk management. Among the participants were MML personnel, specialists from state entities, officials from the district municipalities of the province of Lima, representatives of academia and civil society organizations such as youth organizations, NGOs, among others.

Regarding the measures of the various actors for the comprehensive management of climate change in the province of Lima, the Matrix of Objectives and Strategic Actions of the PLCC was sent to the different institutions, who after reviewing the actions included in the scope of their powers and /or add actions that were not being considered, gave their

compliance and adhered to the PLCC. This in order to work jointly with the MML in the comprehensive management of climate change in the province of Lima.

1.4.1.4. Definition of actions

The fourth phase consisted of defining PLCC actions through a process of prioritizing actions. The long list of actions version of the plan was reviewed with the objective of selecting the actions that, in the context of the city, will provide the greatest benefits for climate change management. This process is necessary, since not all actions can be executed at the same time due to different factors such as the lack of availability of human or budgetary resources; in addition to allowing the identification of those actions with greater viability and impact.

The tool used for the analysis of prioritization of actions was the Action Selection and Prioritization tool - ASAP[16] (for its acronym in English), which considered 3 prioritization criteria: primary benefits, co-benefits and feasibility. To define these criteria, there was the participation and contributions of representatives of various areas of the municipality.

The first is the primary benefits criterion (risk reduction and emissions reduction), the second is the co-benefits criterion (air quality; physical health and well-being; habitat and green spaces; jobs, income and poverty; public services; involvement of actors; and social

justice) and the third is the feasibility criterion (city authority; alignment with city policies; political and social acceptability; and secured source of financing). Then, the goals, actions and co-benefits were written in greater detail and a virtual workshop was held in September 2020 that was attended by actors internal and external to the MML, who were able to learn about the vision for 2030, the objectives of the Plan, strategic actions and inclusion approach for climate action, proposed for the PLCC. At the same time, it allowed the SEACC team to collect contributions and recommendations from the participants on the proposed measures.

1.4.1.5. Plan formulation

The fifth phase consisted of the final compilation of the document by the Deputy Management of Environmental Strategy and Climate Change, based on all the information produced throughout the process and the contributions of the participatory process. Although the process of formulating the Plan was not totally linear, at this stage all the information from the chapters generated was consolidated, after the participatory process implemented by

[16] Excel-based software tool developed by C40. Documents action information from climate action plans and provides graphical results to support the decision-making process by comparing the benefits and challenges of the action. This tool was developed to help C40 cities prioritize climate actions as part of the Climate Action Planning Technical Assistance Program process.

the MML for the construction of the Plan, in order to include in the first version all the contributions of the actors who participated in the process.

1.4.1.6. Plan Review

The sixth phase consisted of the review of the PLCC at the internal and external level of the Municipality of Lima. Internally, the Plan was validated with management and deputy management, decentralized public organizations and municipal companies. While, at an external level, it was sent for reviews by MINAM and C40. After the review process was concluded, the latest version of the PLCC was presented for pre-publication and the draft ordinance that declares the Development of Actions against Climate Change of Metropolitan Interest and approves the Local Climate Change Plan of the Province of Lima 2021- 2030 that will approve it, which was published on the MML transparency page to receive opinions from the general public. Finally, after the pre-publication process, a consolidated version of the draft ordinance focused on climate management in the province of Lima is presented to the Metropolitan Council, including the approval of the aforementioned PLCC.

1.4.2. Plan implementation process

The implementation process of the plan is described in chapter 4, this consists of a series of governance, mitigation and adaptation objectives, for which specific actions are established, under a vision of the city to 2030. This stage will begin after the approval of the plan by the MML.

1.4.3. Plan validation

The validation of the Plan has been constant during the different stages of the formulation process, due to the participatory approach for its design and implementation. This was carried out through the participatory process implemented with various actors internal and external to the MML, taking into account the involvement of specific actors, such as the Environmental Council of Girls and Boys, Youth Organizations and district municipalities.

The types of actors involved are listed below.[17] in the participatory process for the formulation of the PLCC:

Internal actors:

- Municipal managements;
- Decentralized Public Organizations;
- Municipal Companies
- Metropolitan Council

[17] The detailed description of the social actors of the city of Lima for the management of climate change is presented in the Climate Governance subsection of this plan.

External actors:

- State Entities such as the Ministry of the Environment, Ministry of Transportation and Communications, Ministry of Energy and Mines, SENAMHI, among others; ● District Municipalities of Metropolitan Lima;
- Academia;
- Civil society;
- NGOs;
- Youth Organizations;
- Environmental Council for Girls and Boys;
- Metropolitan Environmental Commission – CAMET

As a result of the participatory process**,** In 2019, 12 work meetings were held between the SEACC technical team, municipal management, municipal companies, decentralized public organizations, municipal companies and councilors, as well as with actors external to the Municipality. To see the details of the activities for 2019, review Table 1 of the **Annex 1. PLCC Participatory Process**

While in 2020, 22 work meetings, workshops and conferences were held virtually given the context of COVID-19, among the technical team, with actors internal and external to the Municipality. To see the details of the activities for the year 2020, review Table 2 of the **Annex 1. PLCC Participatory Process**

Finally, as part of this process, 8 meetings of the Metropolitan Technical Group on Climate Change and Water Resources were held, and 4 meetings of the Metropolitan Environmental Commission - CAMET between 2019 and 2020. The participatory process aims to build the PLCC under a participatory approach. which means integrating the needs, visions and perceptions of the public and private social actors of the city of Lima.

1.5. Climate governance

The principle of climate governance establishes that public processes and policies for adaptation and mitigation to climate change are built so that the effective participation of all public and private actors in decision-making, conflict management and consensus building is possible. , based on clearly defined responsibilities, goals and objectives at all levels of government (LMCC, 2018).

The MML has a decisive role in achieving the goals of reducing GHG emissions, reducing the vulnerability of the population, increasing resilience and improving adaptation capacities, which is why it must play an articulating role among all social actors in the city. ; given that, for effective climate governance, the particularities of the province of Lima must be considered, which is characterized by its geographical extension and the large number of district municipalities that make up its political organization. Likewise, for effective climate management, the public sector, the private sector, academia and civil society must be involved, with emphasis on girls, boys and youth.

Added to these characteristics is that there is a diversity of state entities, at different levels of government, that have functions in key sectors, where measures related to climate change must be designed and implemented, such as the decarbonization of the electrical grid, the optimization of energy use in buildings, improving comprehensive solid waste management, increasing the resilience of water and sanitation systems, comprehensive disaster risk management or improving the resilience of natural capital.

In accordance with the aforementioned measures and municipal powers, it is identified that the MML has limited functions for the design and implementation of policies and actions in the sectors considered key, such as the production and distribution of energy and electricity or household water security (water services). water and sanitation). For example, the decarbonization of the electrical grid depends on national bodies such as the Ministry of Energy and Mines (MINEM), which in turn has the Supervisory Body for Investment in Energy and Mining (OSINERGMIN) as a regulatory entity at the sector level. While, to improve the resilience of water and sanitation systems, the person in charge is the Ministry

of Housing, Construction and Sanitation (MVCS), and in turn, the National Superintendence of Water and Sanitation Services (SUNASS) supervises the sector. .

At the moment[18], the Metropolitan Municipality of Lima has different spaces for environmental governance in the province of Lima. One of these spaces is the Metropolitan Environmental Commission (CAMET), which is responsible for the coordination and agreement of the National Environmental Policy and promotes dialogue and agreement between local actors, the Ministry of the Environment (MINAM) and other entities. competent. Likewise, it channels multi-sectoral environmental proposals, and monitors and evaluates them. The creation of formal spaces such as CAMET are a strength for climate governance since they articulate local actors and allow them to work in a joint and institutionalized manner.

Another governance space is the Children's Environmental Council, which is made up of eleven girls and boys, between 7 and 11 years old, who live in Cercado de Lima. Its objective is to support environmental management through its advice and advice on environmental issues that affect the district, developing proposals that contribute to improving the environmental quality of Lima. For the formulation process of the PLCC, we worked with the children through this space, where they had the opportunity to provide their perceptions about the problem of climate change, proposed solutions and their vision of the desired future for the city.

In the case of youth, a participatory process was carried out aimed especially at youth organizations with initiatives in the environment, climate and sustainability sector. Spaces for dialogue and synergy between youth civil society organizations were implemented, which made it possible to generate and collect their contributions to the PLCC.

[18] In April 2020, the Metropolitan Environmental Management System (SMGA) of the province of Lima was updated, the most important framework instrument for the management of the environment of the capital city, which was formulated based on the participation of various actors from the commune. Lima, represented in the Metropolitan Environmental Commission (CAMET).

1.5.1. Role of the Metropolitan Municipality of Lima

The Metropolitan Municipality of Lima is the local government body that exercises exclusive jurisdiction over the province of Lima. It promotes the adequate provision of local public services and comprehensive, sustainable and harmonious development in its territorial constituency. In accordance with the provisions of Law No. 27972, Organic Law of Municipalities, it has a special regime to perform functions as a regional government and a provincial municipality, both within the province of Lima. As part of its functions as a provincial municipality, it is in charge of the Cercado de Lima district.

The LMCC indicates that regional and local governments, within the framework of their powers and functions, are responsible for:

• Execute national public policies on climate change and design, monitor, evaluate and redesign regional strategies on climate change. • Incorporate mitigation and adaptation measures to climate change in its Territorial Plan, Regional and Local Concerted Development Plan, Institutional Strategic Plan, Institutional Operational Plan, Budget Programs and investment instruments.

• Report to the Ministry of the Environment the status of execution of public policies, regional strategies, and mitigation and adaptation measures to climate change incorporated in its planning instruments.

• Promote the development of integrated vulnerability and adaptation studies for the identification of vulnerable areas, as well as scientific research and technological development for mitigation and adaptation to climate change, considering the traditional knowledge of indigenous or native peoples.

• Design, execute, monitor and evaluate measures and projects to develop the capacity to adapt to the impacts of climate change and reduce vulnerability, prioritizing populations in vulnerable situations.

• Develop institutional capacities in concepts and processes related to climate change and mitigation and adaptation measures to climate change • Promote the informed participation of citizens, particularly populations in vulnerable situations, such as women and indigenous or native peoples, in the comprehensive management of climate change, aimed at strengthening climate governance and sustainable development in harmony with nature.

Likewise, the LMCC Regulations indicate that local governments have the following functions related to the comprehensive management of climate change:

- Prepare, approve, implement, monitor, evaluate and update your Local Climate Change Plan, in accordance with the principles and approaches of the LMCC.
- Incorporate adaptation and mitigation measures defined in the PLCC in their plans, budget programs and investment projects.
- Guarantee effective, timely and continuous participatory spaces in the formulation and updating processes of your PLCC.

- Track and evaluate the level of progress in the implementation of your PLCC.
- Other functions that derive from compliance with the LMCC Regulations.

1.5.1.1. City Services and Environmental Management Management

The GSCGA is the line body that exercises the functions of operational supervision of the provision of services to the city.[19] and is responsible for proposing the policy, agenda, environmental plans, education and citizen participation in environmental management; In addition, it issues a technical environmental opinion that corresponds to the MML.

In coordination with the CAMET and the respective Metropolitan Technical Groups (GTM),

the GSCGA monitors the implementation and performance of the Metropolitan Environmental Management System (SMGA), which are reported annually to the MINAM. Likewise, annually, it submits an environmental management report that evaluates compliance with the MML in environmental matters, which is presented to CAMET and published in the Metropolitan Environmental Information System (SMIA).

It is made up of three sub-managements: Sub-Management of City Services, Sub-Management of Environmental Management and the Sub-Management of Environmental Strategy and Climate Change.

1.5.1.2. Deputy Directorate of Environmental Strategy and Climate Change

The Environmental Strategy and Climate Change Sub-Management (SEACC) is the organic unit responsible for proposing the policy, the agenda, the environmental plans, the formation and operation of the CAMET, coordinating the operation of the SGMA, generating participatory spaces that promote environmental governance ; and has been designated to coordinate, formulate and implement the PLCC.

The SEACC's functions in climate change include: coordinating, conducting and supervising the implementation of climate change plans; propose, design and promote the development of mitigation and adaptation initiatives to climate change; and generate periodic information on this matter that feeds the SMIA.

Through Official Letter No. 61- 2020 – MML/ GSCGA, the MML designated the SEACC as the MML's Climate Change Focal Point vis-à-vis the MINAM, other competent authorities and non-state actors, as established in the LMCC regulations.

1.5.1.3. MML areas involved in climate change management

- Metropolitan Council: Regulatory and supervisory body. It is made up of 39 MML councilors. Its functions include approving council agreements and municipal ordinances and the annual budget.

[19] Spublic cleaning service, maintenance of urban infrastructure, ornamental lighting and lighting artistic, ornament of the Cercado de Lima and the metropolitan roads.

- Metropolitan Mayor's Office: Executive body of the Local Government. The mayor is the legal representative of the Municipality and its highest administrative authority. Issues administrative regulations, mayoral decrees and mayoral resolutions.

- Managements, Deputy Managements, Decentralized Public Organizations, Municipal Companies, among others.

    - Administration Management: responsible for administrative support through the administration of human and logistical resources, the provision of general services and directing digital government and innovation.

- Economic Development Management: responsible for promoting the innovation system within the productive base and business fabric of the province of Lima.
- Social Development Management: responsible for social development policies and actions, mapping of vulnerable populations and strategies for their care.
- Urban Development Management: responsible for spatial planning and development in Metropolitan Lima.
- Finance Management: responsible for the administration of the financial resources of the municipal system within the scope of its jurisdiction, in accordance with the Organic Law of Municipalities and other applicable legal provisions.
- Urban Mobility Management: regulation of urban mobility, transit, freight transportation, minor vehicles and special use of roads and other infrastructure that allow the development of mobility in the province of Lima.
- Neighborhood Participation Management: responsible for strengthening ties between neighbors and the Municipality, facilitating the empowerment of society.
- Planning Management: responsible for leading and supervising planning, organization and modernization processes, multi-year investment programming, and leading international technical cooperation strategies.
- Private Investment Promotion Management: responsible for promoting private investment, establishing strategic alliances with the national government, regional and local governments, private investment and civil society, in assets, companies, services and works of the MML.
- Disaster Risk Management Management: responsible for conducting and supervising civil defense and disaster risk management processes in the jurisdiction of the province of Lima, in accordance with the regulations that regulate the matter.

- Metropolitan Lima Regional Government Program: deconcentrated body whose purpose is to coordinate, organize, create and manage the necessary conditions for the efficient development of the transfer process and the orderly reception of sectoral functions and competencies from the National Government to the MML.

The following sub-management areas have been identified as relevant areas for climate management:

- Regional Agrarian Deputy Management, for its relationship in agricultural projects.

- Regional Deputy Management of Economic Development, for its relationship in the sectors of Industry, Commerce, Tourism, Crafts, Fisheries, Mining, Energy and Hydrocarbons.
- Regional Deputy Management of Natural Resources and Environment, because it is the line body in charge of attending to specific functions regarding management in basins, water resources and protected natural areas.

- Costa Verde Project Authority - APCV: responsible for promoting, ordering and supervising the comprehensive and sustained development of the Costa Verde; and organize and direct the set of planning actions for urban, tourist, cultural,

recreational and environmental development in the Costa Verde area.

- Municipal Authority of the Pantanos de Villa - PROHVILLA: responsible for the management of the Special Regulation Zone of the Pantanos de Villa, which has the purpose of conserving the protected natural area of the Pantanos de Villa. In addition, it carries out environmental surveillance work in the Special Regulation Area (ZRE) of the Pantanos de Villa.

- Municipal Toll Administrator Company of Lima - EMAPE: responsible for the construction, remodeling, conservation, operation and administration of highways, roads and other rapid transit routes (urban, suburban or interurban), including their access roads, bridges, crossings. unevenness, service areas, recreation areas, decoration and annexed areas. Responsible for the collection and administration of the toll system.

- Municipal Real Estate Company of Lima - EMILIMA: responsible for the administration of properties owned by the Municipality of Lima, formulates and executes investment projects related to municipal properties. Designs, executes and evaluates policies, standards, plans, programs and projects within the scope of its competence, aimed at urban development, within the scope of its competence.

- Metropolitan Planning Institute - IMP: responsible for formulating and evaluating the different municipal plans, proposing the investment program and medium and long-term financial strategy, convening economic and social agents from the public and private sectors in local development processes , formulates investment projects in the Metropolitan area, manages the Investment Project Bank of Metropolitan Lima.

- Municipal Program for the Recovery of the Historic Center of Lima - PROLIMA: responsible for promoting the recovery of the Historic Center of Lima (CHL), declared Cultural Heritage of the Nation and inscribed on the UNESCO World Heritage list, the entity in charge of implementing the actions of the PLCC in the area of the CHL.

- Lima Parks Service - SERPAR: responsible for managing the metropolitan and zonal parks of Lima.

- Metropolitan Solidarity System - SISOL: responsible for providing specialized health services to the population, at affordable prices and in a timely manner; to improve the quality of life of people in the province of Lima.

For more information about the Areas involved in Climate Change Management, contact **Annex 2. PLCC Human Resources Chart**, where the human resources available to the MML, Municipal Companies and Decentralized Public Organizations for the implementation of the PLCC are presented in a disaggregated manner. The number of actors involved is established, as well as the function of each one.

1.5.1.4. City Environmental Management Instances

In the province of Lima there are institutionalized environmental management bodies that were used for the formulation of the PLCC, which will be maintained during the implementation stage, such as the Metropolitan Environmental Management System, the Metropolitan Environmental Commission (CAMET), the Metropolitan Technical Groups and the Environmental Council of Girls and Boys. There are also other mechanisms that contribute to environmental governance such as alliances with academia, NGOs, civil society and the private sector.

1.5.1.4.1. Metropolitan Environmental Management System (SMGA)

The SMGA is the set and interaction of principles, norms, processes, instruments, spaces for consultation (Metropolitan Environmental Commission, among others) and people involved in the various systems and thematic areas.[20] of the National Environmental Management System (SNGA) in the province of Lima, likewise, establishes its articulation with other systems such as the National Forestry and Wildlife Management System (SINAFOR) and the National Disaster Risk Management System (SINAGERD).

The City Services and Environmental Management Management (GSCGA) is the executive, supervisory and supervisory body of the Metropolitan Environmental Management System, which was created by the Metropolitan Municipality of Lima to organize the governability and environmental governance of the territory, as well as articulate the set of actors with competencies and
environmental responsibilities.

[20] The thematic areas refer to environmental information management, environmental impact assessment, environmental assessment and oversight, conservation of biological diversity, water resources management, comprehensive management against climate change, environmental territorial planning, comprehensive solid waste management. , air quality management, management of the fight against desertification and drought, and environmental education and citizenship.

Figure 2. Representation of the Metropolitan Environmental Management System (SMGA). Source: (MML,2020).

1.5.1.4.2. Metropolitan Environmental Commission (CAMET)

Within the framework of the SMGA, the Metropolitan Environmental Commission (CAMET) is the body in charge of coordinating and agreeing on the National Environmental Policy in the province of Lima. It is made up of 26 representatives of the public sector, civil society, and the private sector, convened by the GSCGA and appointed by mayoral decree, their participation is ad honorem, and their term of office is 2 years.
CAMET has promoted the creation of the Metropolitan Technical Groups (GTM)[21], which are consultation spaces for local environmental management. These are made up of specialists who propose action mechanisms for the design and evaluation of environmental management policies, standards and instruments. The Metropolitan Technical Group on Climate Change and Water Resources has been an important space in the construction of the PLCC, since the advances have been presented and validated by the members of this group. It is important to maintain the creation of a technical group related to climate change for the monitoring and evaluation of the PLCC.

1.5.1.4.3. Metropolitan Environmental Information System (SMIA) Information transparency is a pillar of municipal governance because it legitimizes government agencies as democratic systems, in which the population has free access to different types of documentation (instruments, standards, plans, statistics, etc.)

[21] Among the groups created for the CAMET 2019-2020 period are the GTM of Environmental Governance and Citizenship; GTM of Climate Change and Water Resources; GTM of Ecosystem Management and Conservation; GTM of Solid Waste Management; and GTM for Air Quality Management.

The SMIA constitutes a technological, institutional, technical and human integration network to facilitate the systematization, access, dissemination and distribution of environmental information, as well as to establish and guarantee a fluid exchange of information for decision-making and management processes. environmental management in the jurisdictional area of the province of Lima. It is governed by the Law of Transparency and Access to Public Information[22] and its amendments, and what is stipulated by Municipal Ordinance 2256[23]. The SMIA is formed as a mechanism of the MML for climate governance and therefore contributes to the comprehensive management of climate change.

To make environmental content visible, the SMIA has been made available to citizens; for the dissemination of this system, the Lima Environmental Facebook page is used, where the informative notes and featured documents are republished; In addition to this, a monthly SMIA newsletter is sent with a summary of the platform's contents. The distribution of information on the web portal is governed and allows interoperability under the framework of the National Environmental Information System (SINIA), detailing the same 13 environmental topics and including climate change.

1.5.1.4.4. Promotion of environmental research

The promotion of research on environmental issues is necessary to generate information, knowledge and capabilities for the benefit of the population and the environment. This action is in charge of the Sub-Management of Environmental Strategy and Climate Change (SEACC) of the GSCGA. Which has been promoting research in environmental matters through the dissemination of a list of 69 environmental needs and problems. This aims to generate the participation of universities, research institutes, specialized entities and citizens; as well as, positioning these problems as axes of research among students, teachers and researchers.

1.5.2. Actors for the comprehensive management of climate change in the province of Lima

The Framework Law on Climate Change and its regulations establish, through the principle of governance, the participation of state and non-state actors for the comprehensive management of climate change. In Peru, the Ministry of the Environment (MINAM) is the national authority on climate change, guiding and conducting its comprehensive management within the framework of its powers. This includes the participation of other state entities, competent in the matter, to which specific functions are designated, such as ministries, attached entities, regional governments and local governments. In this context, civil society, academia, the private sector, youth, girls and boys are also recognized as relevant agents for climate governance.

[22] Law No. 27806, Law on Transparency and Access to Public Information.

[23] Ordinance that updates the Metropolitan Environmental Management System (SMGA) of the province of Lima

1.5.2.1. State actors

1.5.2.1.1. national authority

The MINAM is the national authority on climate change and the technical regulatory authority at the national level on this matter; Among its functions are coordination with the Climate Change Focal Points for the formulation, implementation, monitoring, evaluation, reporting and updating of its policies, strategies, plans, programs and projects that incorporate adaptation and mitigation measures that contribute to comprehensive management. of climate change and the implementation of the NDCs. Regarding local plans on climate change, it is responsible for the preparation and approval, in coordination with sectoral authorities, regional and local governments, of the guidelines for the formulation of ERCC and PLCC.

1.5.2.1.2. Sectoral authorities

- Local Water Administration Chillón - Rímac - Lurín - ALA: Administrative entity of the water resources of the Chillón, Rímac and Lurín rivers, its functions include support for the AAA and actions for the supervision, control, surveillance and inspection that ensure the sustainable use, conservation and protection of the quality of water resources.

- Cañete – Fortaleza Water Administrative Authority - AAA: Decentralized body at the national level that addresses the water needs of the Ancash, Callao, Junín and Lima regions. In Lima it orients its management to the supply of drinking water due to the size of the population and the local industries.

- Autonomous Authority of the Electric Mass Transportation System of Lima and Callao: Entity – AATE: Entity in charge of leading the consolidation of the basic network of the Lima and Callao Metro. This transportation method under construction promotes electric transportation. It will join the ATU.

- Urban Transportation Authority - ATU: Entity that is in charge of planning, regulating and supervising the land transportation service in the capital with the objective of guaranteeing the sustainability of this mode of transportation.

- National Water Authority - ANA: Governing body and highest technical regulatory authority of the National Water Resources Management System, which is part of the National Environmental Management System.

- National Center for Disaster Risk Estimation, Prevention and Reduction – CENEPRED: Executing public agency with the functions of regulating, managing and technically assisting entities in the estimation, prevention and reduction of disaster risk and reconstruction.

• Interregional Council of Cuenca Chillón – Rímac – Lurín: Permanent body of the ANA, which aims to participate in planning,

coordination and agreement on the sustainable use of water resources in its area.

• Regional Directorate of Education of Metropolitan Lima - DRELM: Body that aims to apply, execute and manage the national educational policy issued by the MINEDU and evaluate its implementation in the jurisdiction of Metropolitan Lima.
• National Institute of Civil Defense - INDECI: Organization that seeks an optimal response of society in cases of disaster: this includes supervising the care of affected people and coordinating with the responsible entities the actions required to address emergencies.

• The General Directorate of Energy Efficiency is the line body in charge of proposing and evaluating energy efficiency policies, non-conventional renewable energies and promoting the formation of a culture of rational and efficient use of energy, as well as conducting energy planning.

• Ministry of Defense – Directorate of Hydrography and Navigation: administrative, operating and research entity of activities related to environmental sciences in the aquatic field in order to contribute to national development.
• Ministry of Energy and Mines - MINEM: Governing body of the energy and mining sector of Peru, it exercises powers in matters of energy, which includes electricity, hydrocarbons and mining.

• Ministry of Production - PRODUCE: Governing body of national productive policy, in charge of formulating, approving, executing and supervising all levels of production, industry, manufacturing and fishing. It is also responsible for waste management policies in the manufacturing sector.

• Ministry of Transportation and Communications - MTC: Governing body for the policies and development of the country's transportation systems and communications and telecommunications infrastructure.

• Ministry of Housing, Construction and Sanitation - MVCS: Ministry responsible for the housing, construction, sanitation, urban planning and urban development sectors within the territory of Peru. Responsible for promoting sustainable cities and housing, and ensuring energy efficiency in new buildings in the city. It is also responsible for waste management policies in the construction sector.

• Lima Drinking Water and Sewerage Service- SEDAPAL. State company in charge of providing drinking water and sewage to the urban sector of Metropolitan Lima. It is part of the Metropolitan Technical Group for Climate Change and Water Resources of CAMET.

• National Service of Natural Areas Protected by the State - SERNANP: Specialized

technical public body, in charge of directing and establishing the

technical and administrative criteria for the conservation of Protected Natural Areas, aims to conserve the biodiversity of Peru.

- National Service of Meteorology and Hydrology of Peru - SENAMHI: Entity in charge of generating and providing meteorological, hydrological and climatic information and knowledge in a reliable, timely and accessible manner.

- National Forestry and Wildlife Service - SERFOR: Specialized technical public body, it is the national forestry and wildlife authority. Promotes sustainable, inclusive and competitive forestry and wildlife management, which allows us to rise to the challenges generated by climate change and the pressure exerted on the world's forests.

1.5.2.1.3. Regional and local authorities

- Regional governments: regional authorities must develop climate change strategies based on the Framework Law on Climate Change, such as issuing recommendations and contributions in the formulation and updating stage of local climate change plans within their jurisdiction. As well as annually consolidate the reports on the level of progress in the implementation of the PLCC of the provincial and district governments, to send them to the MINAM.

- Local governments: provincial and district municipalities, based on the principles and approaches established in the LMCC, must develop, approve, implement, monitor, evaluate and update their PLCC. Incorporate adaptation and mitigation measures defined in the PLCC in your plans, budget programs and public investment projects. Request the regional government or provincial municipal government of your jurisdiction, as appropriate, for recommendations and input in the formulation and updating stage of your PLCC. Guarantee effective, timely and continuous participatory spaces, in the processes mentioned above. And monitor and evaluate the level of progress in the implementation of your PLCC.

1.5.2.2. Non-state actors

Non-state actors are part of climate governance, one of the main actors that must be involved for the city of Lima is civil society, which is understood as the set of voluntary and social civic organizations and institutions, related mainly to exchange climate change, as well as those who, due to their role, contribute to the effective implementation of the PLCC, are part of this group, and work in prioritized sectors for climate change:

- Non-governmental organizations
- Public-private platforms
- Professional unions

- Youth Organizations
- Social organizations that represent people with disabilities

Other non-state actors are:

- Academy: is made up of the various cultural or educational institutions that exist in the country, with emphasis on the city of Lima, and that contribute to the management of climate change in the educational field.
- International cooperation: it is made up of the agents that provide support at the international level to different actors such as governments of different countries, companies, NGOs, among the main ones, in order to promote economic and/or social development, and in this case, to efforts for climate action in the city.
- Private sector: it is made up of large and small companies, which have influence in the city due to their economic activities, and which are currently not involved in the management of climate change. It also includes its business unions.

1.6. Characteristics of the study area

1.6.1. Physical and geographical

Lima is the capital of Peru, with a territory of 2,641.89 km^2, is the largest, most extensive and most populated metropolitan area in the country. Geographically, it is located within the Lima region, on the central coast of the country, on the western slope of the Andes Mountains, facing the Pacific Ocean, at an altitude that varies between 0 and 850 meters above sea level. sea.

The geographical limits of the province of Lima are:

- To the north with the province of Huaral
- To the south with the province of Cañete
- To the east with the provinces of Canta and Huarochirí
- To the west with the constitutional province of Callao and the Pacific Ocean

Figure 3. Geographic setting.
Source: (IDOM, 2020)

At the political level, it has a special regime with powers of both local government (district and provincial) and regional government. Administratively it is divided into 43

districts, not including the Constitutional Province of Callao, with which it is conurbated. Due to their extension and size, the different districts of the city are organized into four group zones: center, south, east and north.

Figure 4. District division.
Source: (IDOM, 2020).

On a physical level, it is characterized by a relatively flat topography, although with important changes in its relief towards the east, appearing areas of foothills and small mountains. Most of the urban footprint of Lima and Callao sits on the coast, between the mean sea level on the Pacific coast and the low foothills of the Andean fortress, at an altitude of about 500 meters above sea level. Towards the east it reaches 2,700 meters

above sea level. in the mountain range of the Chillón River basin. The structural foundation of the Ande mountain ranges clt conditions its geomorphological configuration, making the city have a stepped profile with well-defined jumps (IMP, 2014).

The metropolitan area is characterized by the following components:

- The coastline is generally formed on terraces of low and medium height - such as those of Chorrillos Bay - and, in some cases, a coastline almost at sea level predominates - as on the southern beaches.
- The desert area is made up of a succession of pampas, hills and platforms that begin on the coastal beaches and extend to the east to the limit of the Andean foothills. This strip is quite narrow and not very rugged, with an average width of 10 km.
- The valleys formed by the waters of the Chillón, Rímac and Lurín rivers that have variable flow; and towards the bottom of the valleys, and streams that remain dry periodically and that react in the rainy season (December to March)

violently with torrential flows of mud and sludge called huaycos[24]. They also include areas of hills, oases of fog, water attractors and oxygen producers, located in the districts of Comas, Villa María del Triunfo, Pachacámac, Lurín and Rímac. Furthermore, the urban area is located in two ecoregions: the ecoregion of the cold sea of the Peruvian current and the ecoregion of the Pacific desert. The first and its low temperatures provide a sea that has one of the largest schools of fish in the world and great marine biological diversity.

Figure 5. Geomorphological units.
Source: (IMP, 2014).

At the level of hydrographic basins, the city has river plains associated with the final reaches of the Lurín, Rímac and Chillón rivers. The structure of these basins has fractures, faulting and folding due to the continental drift of the South American tectonic plate on the bottom of the Pacific Ocean. This tectonic process has led to the formation of a subduction trench on the ocean floor (200 km west of the Lima coast), parallel to the South American continent, and which is a source of seismic energy release.

The main characteristics of the rivers that cross the city are:

[24] Huayco is also called mass movement, it is a displacement of light to moderate magnitudes of small or large masses of water with a mixture of soils, mud and rocks. Extracted from the National Disaster Risk Management Plan (PLANAGERD) 2014-2021, approved by DS Nº 034-2014-PCM

- The Rímac River begins to open from the Chaclacayo sector until reaching the vicinity of the Army Bridge, where an extensive flood plain has formed. The Rímac in times of exceptional floods increases its erosive power and in unprotected areas landslides can occur.
- The Chillón River has formed an extensive flood plain from the sugar mill sector to its mouth. Part of the riverbed has been channeled with the construction of dams on

both banks.

- The Lurín River receives contributions from its tributary streams, where exceptional flows are generated. In this sector, low terraces are formed that are very susceptible to flooding. Continuing downstream, dry streams only active during the El Niño phenomenon are observed on both banks.

Figure 6. Relief and basin structure.
Source: (IMP, 2011).

1.6.2. Climate

The climate of the Metropolitan Lima region is characterized by being semi-warm-arid with an average annual temperature of 18ºC, with precipitation deficiency in all seasons of the year and humid (SENAMHI[25], 1988) due mainly to the presence of the South Pacific Anticyclone (APS), which exerts a strong subsidence on the west coast of South America (north coast of Chile and south and central coast of Peru), being the factor that inhibits the formation of cloudiness. in this region, giving it the category of one of the driest regions in the world. The historical record of trends is shown below.

[25] National Service of Meteorology and Hydrology of Peru: It is the entity officially delegated to generate and update the national climate scenarios, as established by the Regulation of the Climate Change Framework Law published in the newspaper "El Peruano" on 12/31/2019 and approved with Supreme Decree 013-2019-MINAM.

of temperature and precipitation variation, which constitutes a baseline for the modeling of future climate change scenarios.

1.6.2.1. Temperature

The National Meteorology and Hydrology Service of Peru (SENAMHI) has meteorological stations distributed nationwide, which allows analyzing the trends of the annual minimum and maximum temperature variables. According to the reference climatology, for the period 1971-200 (SENAMHI, 2015) the maximum annual temperature (Figure 7) fluctuates between 19ºC and 25ºC in the province of Lima. This temperature has the highest values on the coast and decreases progressively as altitude increases. Regarding minimum annual temperatures (Figure 8), they range between 8ºC and 17ºC. The highest values are observed on the coast due to the thermoregulatory effect of the sea and decrease with altitude, as the distance from the sea increases.

Figure 7. Maximum annual temperature.
Source: (IDOM, 2020).

Figure 8. Minimum annual temperature.
Source: (IDOM, 2020).

In general, temperatures have a sinusoidal character[26], which differentiate the months from June to September with the lowest values (around 12ºC for the minimum temperature; around 16ºC for the average temperature; and around 21ºC for the maximum temperature) and the months from December to April with the highest ones (around 18ºC for the minimum temperature; around 22ºC for the average temperature; and around 27ºC for the maximum temperature).

1.6.2.2. Precipitation

With regard to precipitation, the four previously mentioned factors (Humboldt Current, Andes Mountains, topography and the continental effect) determine the desert condition of Metropolitan Lima, which in turn is affected by precipitation coming from the basins. high. The total annual precipitation is less than 50 mm and increases with altitude, reaching maximum values of 178 mm (SENAMHI, 2015).

[26] Sinusoidal: Which can be represented graphically by a sinusoid, the latter is a flat curve obtained from the representation in Cartesian coordinates of the sine function.

Figure 9. Average annual total precipitation.
Source: (IDOM, 2020)

1.6.3. Water resources

Regarding trends, the Ñaña station is the only one that presents a trend in its record, which is characterized by increasing 3.8 mm per decade. The Mann-Kendall test does not show a trend for the rest of the stations in Metropolitan Lima.

Lima is located in three basins: Chillón, Rímac and Lurín. Among the most important characteristics of the surface hydrographic network, the irregular hydrological regime stands out, with the flows of the three main rivers being very variable, becoming scarce (in the Rímac river) or almost non-existent (in the Chillón and Lurín) at the time. dry season between June and November (SENAMHI, 2016). Figure 10 shows the series of average monthly flows for the period 1970-2015 in these three rivers.

Figure 10. Monthly variation in the flow of the Chillón, Rímac and Lurín rivers. Source: (SENAMHI, 2016).

Rímac River Basin: The Rímac River basin is located between parallels 11°30' and 12°15' south latitude, 76° and 77° west longitude. Its area is 3,240.6 km2. It is politically located in the jurisdiction of the department of Lima and to a lesser extent in the department of Junín, falling within the provinces of Lima, Huarochirí and Yauli. The average flow is 30 m^3/s, and it has its own regulated system and a transfer from Mantaro with an average flow of 6.8 m^3/s.

Chillón River Basin: The Chillón River basin is located between parallels 11°15' and 12°00' south latitude and 76° and 77° west longitude. Its total area is 2,181.5 km2. It is located within the department of Lima and in the constitutional province of Callao, falling within the provinces of Lima (eight districts) and Canta (seven districts).

Lurín River Basin: The Lurín River basin is located between parallels 11°45' and 12°15' south latitude, 76° and 77° west longitude. Its area is 1,568.5 km^2 and its average flow is 5.2 m^3/s.

Figure 11. Location and limits of the Chillón Rímac Lurín Interregional Basin Water Resources Council.
Source: (Water Observatory, 2018).

The main sources of water for Lima and Callao are the Rímac River and the transfer from the Mantaro River basin, where the Marcapomacocha lagoons and/or reservoirs are located. Lima also has underground water sources. The supply is made up of water reserves stored in the aquifer matrix, and dynamic, natural or artificial recharge (SEDAPAL, 2014). Figure 12 shows the drinking water supply system of the city of Lima delimited by each water source.

Figure 12. Water sources for Lima.
Source: (SEDAPAL, 2018)

However, despite their importance and the ecosystem services they provide to citizens, rivers have suffered a significant decrease in the quality and volume of their waters, caused mainly by anthropogenic polluting agents. This degradation has an impact on an increase in natural phenomena such as droughts, floods, erosion and sediment transport; taking into consideration the inventory carried out by INGEMMET until 2018 and the SENAMHI-SEDAPAL study from 2016, where a total of 777 dangers of natural disasters that occur in the basins are evident. A clear example of the impact generated is what happened in the river
Rímac, which receives from its source to its mouth into the sea, a total of four hundred and fifty (450) direct waste discharges from miners, industrialists and sewers, without any treatment, this represents an imminent risk to its biodiversity and human health.

This problem has been identified in the area of the main rivers of the Chillón, Rímac and Lurín Interregional Basin Water Resources Council (CRHCI CHIRILU) for which in recent years concerted actions have been carried out between the ANA and other actors such as SEDAPAL, Local Governments, MINSA, User Board, OEFA, MINAM, MINEM and MVCS, with the objective of inventorying and georeferencing the main polluting sources that negatively impact the quality of water resources. The participatory identification of polluting sources was a work led by the ANA and the ALA Chillón Rímac Lurín, carried out in the Chillón (year 2016), Rímac (year 2016) and Lurín (year 2015) rivers.

Table 1 summarizes the polluting sources identified in the three river basins. Demonstrating that the Rímac River basin is the one with the largest number of polluting sources, mainly pipes connected directly to the riverbed, solid waste and wastewater discharges.

Pollutant source	Gaudy	Rímac	Lurin

Water discharges residual		25	172	8
Water discharges from irrigation		-	-	3
Pipes with direct connection to the channel		7	302	-
WWTP effluent		1	221	4
Dumps of waste solids	**Domestic**	34		20
	Construction	5		4
	Waste organic	-		5
Polluting sources		16	5	23
Mining liabilities			27	
Total		88	727	67

Table 1 Pollutant sources in the Chillón, Rímac and Lurín basins.
Source: (Chillón Rímac Lurín Water Observatory, 2019)

The Lima Drinking Water and Sewerage Service – SEDAPAL S.A is a private law state company entirely owned by the State, which provides drinking water and sewage services in Lima and Callao, and also manages the supply of drinking water to the urban sector of the city. city.

The drinking water system is made up of four major processes: storage, collection, production and distribution, for which SEDAPAL has an infrastructure that is constantly expanding according to the growth of population demand, as well as the areas that are assigned to it. its jurisdictional scope.

Storage: The water resources that SEDAPAL uses to supply the population of Metropolitan Lima and attached areas come from surface and underground water sources. Currently, to satisfy the demand for water in dry seasons, SEDAPAL has built reservoirs, natural reservoirs and/or dams located in the peaks of the central mountain range with the aim of regulating runoff from the Rímac River. The useful storage capacity of these structures is 331 million m^3 and is made up of 19 lagoons and 3 dams strategically located in the Mantaro, Santa Eulalia and San Mateo basins, thus allowing water to be transferred to ensure supply to the city.

Raw water collection: SEDAPAL has three surface water collection systems; two intakes at the La Atarjea Plant (with a collection capacity of 35 m^3/s together), an intake for the Huachipa Plant (with 10 m^3/s) and one for the Chillón Plant.

Drinking water production: During 2018, total production was 729.3 million m^3, coming from the Drinking Water Treatment Plant (PTAP), La Atarjea (72%), PTAP Huachipa (5%), PTAP Chillón (3%) and underground sources (20%).

Distribution system: As of December 2018, SEDAPAL had a length of 14,772 km of drinking water networks, having increased by 412 km in the last 4 years. In order to facilitate the operation of the distribution system, improve service pressures and reduce water losses, SEDAPAL continues with the implementation of the sectorization of drinking water networks, which as of December 2018 reached 168 sectors in operation and of the of which 148 are in a controlled state.

The sewage system includes the processes of collection, treatment and final disposal of wastewater.

Collection system: It is made up of 14,163 km of sewer networks, of which 1,041 km correspond to primary collectors and 13,122 km to secondary collectors. Likewise, the system is complemented by drain pumping chambers, impulsion lines and conduction lines to the wastewater treatment plants.

Sewer networks are affected by toxic elements, discharges of non-Domestic wastewater discharged by businesses and industries, which, when exceeding the maximum admissible values, cause deterioration in the collectors, damage to the treatment infrastructure and blockages, reducing their useful life and generating for SEDAPAL increases in operation and maintenance costs.

For this reason, SEDAPAL permanently carries out maintenance and repair work on the collector network and quality control of non-domestic wastewater discharges.

Wastewater Treatment Plants (WWTP): SEDAPAL has 23 (PTAR), which in 2018 treated a total flow of 22.0 m^3/s, of which 84% correspond to treatment of the Taboada and La Chira WWTPs, and the remaining 16% to the other plants, highlighting the San Bartolo WWTP (with 1.1 m^3/s).

The flow as of December 2018 increased by 1.7 m^3/s compared to the previous year, mainly due to the increase in connections and water consumption in the southern area of Lima, which leads to an increase in wastewater directed to the La Chira WWTP.

SEDAPAL has signed agreements for the reuse of treated wastewater from the San Bartolo, Santa Clara, San Juan de Miraflores, Huáscar, San Bartolo-Balnearios Norte and Sur WWTPs.

Likewise, public entities (such as the Lima Parks Service-SERPAR) and private entities have been reusing treated wastewater for irrigation of parks and gardens, as well as for cultivation projects. It is estimated that 0.80 m^3/s was reused during 2018.

The Carapongo WWTP has 15 photovoltaic (PV) solar panels, which use solar energy to generate electrical energy. The installation of the panels has generated 1,117,375.71 kwh of accumulated energy in 2018, which allowed them to be integrated into the plant's low-voltage network to feed active energy proportionally to the different equipment involved in the wastewater treatment.

1.6.4. Ecosystems

Ecosystems are a “dynamic complex of plant, animal and microorganism communities and their non-living environment that interact as a functional unit”[27] that are part of the natural heritage of the Nation, and since they provide goods and services to the population, they constitute natural capital; Therefore, its use must be sustainable and protected by national, regional and local policies. Within the province of Lima, five ecosystems are identified: coastal hills, coastal wetland, coastal desert, Andean scrubland and riparian ecosystems as represented in Figure 13.

Figure 13. Ecosystem map of the province of Lima.
Source: (MINAM,2018).

[27] Ministerial Resolution No. 440-2018-MINAM. Approve the National Ecosystem Map; Approve the document called "Descriptive Report of the National Ecosystem Map" and Approve the document called "Conceptual Definitions of Ecosystems." Peru.

According to the Regulation of Law No. 30215, Law of Remuneration Mechanisms for Ecosystem Services, ecosystems are important because their proper functioning produces various ecosystem services represented in economic, social and environmental benefits, direct and indirect, for the population. The ecosystems present in the province of Lima provide the following ecosystem services:

- They are the habitat of native and migratory flora and fauna species.
- They provide cultural ecosystem services that encourage the promotion of tourist, cultural and recreational activities.
- They capture carbon and are producers of clean air.
- They provide scientific, cultural, aesthetic, biological, educational information necessary for monitoring and research.

Below, the main natural ecosystems and urban ecosystems that the province of Metropolitan Lima presents are detailed.

1.6.4.1. Coastal hills

It is a coastal desert ecosystem, known as “mist vegetation oasis”, which corresponds to ephemeral xeric plant formations, which include herbaceous plants, with scattered trees in some cases and rich in plant endemisms, which seasonally cover extensive desert areas in the hills. and medium hills exposed to winter fogs. It has a high relative humidity above 80% and the collection of water drops by the bush and tree vegetation, from 100 to nearly 1,000 meters above sea level. (MINAM,2019)

The Province of Lima currently has around 31 thousand hectares of annual hills, distributed in 19 districts from Ancón to Pucusana. These ecosystems are unique and seasonal and green up every year with the arrival of winter, covering the hills of the city. They are recognized as fragile ecosystems by the General Environmental Law[28] and they are part of the MIDAGRI sectoral list of fragile ecosystems (see Figure 14).

The coastal hills are important purifiers of the environment and producers of clean air. The estimated value of the carbon stock in the Lomas de Ancón (Quebrada Inocente) amounts to 8.39 tCO_{2eq}/ha (Cabrera, 2016). Therefore, it is estimated that the total coastal hills in the Province have a storage of around 260,090 tCO_{2eq}, thus contributing to the mitigation of climate change.

These ecosystems are also important because they host a large number of endemic species with a great biodiversity of native and migratory flora and fauna, and a cultural wealth evidenced in the presence of monuments and archaeological remains. Likewise, a large part of this ecosystem is located on peasant communities in the Province of Lima.

Regarding its biodiversity, it is estimated that the richness of its flora is represented by 219 species, grouped into 31 orders and 55 families, of which 85% (170 species) are considered native and 28% (55 species) are considered native. They are endemic, and seventeen species are within the threatened category of wild flora. In

[28] Article 99, Law No. 28611, General Environmental Law.

Regarding the richness of the fauna, it is made up of 94 species, of which 33 are species of birds, 15 are mammals, 10 are reptiles and amphibians, and 36 are invertebrates, including insects, gastropods, arachnids and crustaceans. Of the recorded species, approximately 14% (13 species) are endemic and 16 species are within a threatened category.[29].

The lomas ecosystem has suffered a significant loss of its surface; between 2013 and 2019, a reduction of 680.68 ha was recorded.[30] (of hills). The loss of this surface is a consequence of changes in land use, invasions, land traffic, opening of roads and trails, among others. These actions not only impact the reduction of the ecosystem, they also threaten its biodiversity, affecting the habitat, the landscape, the soil and the generation of ecosystem services to the surrounding populations. On the other hand, they limit the possibility of developing cultural services, including recreational, educational and cultural activities, such as the establishment of interpretation centers, guided tours, scientific research, bird watching or hiking (see Figure 14). .

Figure 14. State of the coastal hills of the province of Lima.
Source: UNDP EbA Lomas Project.

On December 7, 2019, through Supreme Decree N°011-2019-MINAM, the Lomas de Lima System Regional Conservation Area (ACR) was established; which is administered by the Metropolitan Municipality of Lima through the Program

[29] The information on biodiversity of lomas takes as reference the information developed in the technical file of the ACR-Sistema de Lomas de Lima, reports and technical proposals developed by the EbA Lomas Project of the UNDP, the flora guide of the lomas of Lima, the list of species of wild flora and fauna in fragile ecosystems, and the ministerial resolutions, reports and technical sheets of fragile ecosystems developed by the National Forestry and Wildlife Service-SERFOR.

[30] The hill loss information was provided by the EbA Lomas Project of the United Nations Development Program – UNDP.

Regional Government of Metropolitan Lima-PGRLM. The ACR is made up of five coastal hills, located in 7 districts, presenting a total area of 13,475.74 ha, and becoming the first ACR in the city (see Table 2). Its objective is to conserve the lomas ecosystems, their biological diversity and associated cultural heritage, thus guaranteeing the provision of ecosystem services for the benefit of local populations and contributing to the mitigation of climate change.

1	Lomas de Ancón	Ancon	12,166.61
2	Lomas de Carabayllo 1	Carabayllo	228.97
3	Lomas de Carabayllo 2	Carabayllo	198.26
4	Lomas de Amancaes	Rímac, Independencia, San Juan de Lurigancho	253.96
5	Lomas de Villa María	La Molina, Villa María del Triunfo	627.94
Total			**13,475.74**

Table 2 Lomas of the ACR-Lomas de Lima System. Fountain: Own elaboration.

Name	Base Legal	Recognition Date	Extension (Ha)
Lucumus	RMN°0274-2013-MINAGRI	01/08/2013	1597.36
Lurin	RMN°0397-2013-MINAGRI	14/10/2013	1372.07
Pachacamac	RM N°0398-2013-MINAGRI	14/10/2013	4547.69
agreements	RMN°0400-2013-MINAGRI	14/10/2013	993.47
Villa María del Triunfo (*)	RMN°0401-2013-MINAGRI	14/10/2013	691.48
Retamal	RMN°0403-2013-MINAGRI	14/10/2013	656.27
Amancaes (**)	RMN°0404-2013-MINAGRI	14/10/2013	237.45
Panic	RMN°0408-2013-MINAGRI	15/10/2013	705.86
Carabayllo (***)	RMN°0429-2013-MINAGRI	30/10/2013	1767.75
Puquio	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	221.42
Km22	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	77.29
Collic	RDEN°153-2018-MINAG	18/07/2018	369.96

	RI SERFOR-DE		
Payet	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	402.89
Mangomarca	RDE N°153-2018-MINAGRI SERFOR-DE	18/07/2018	516.1
Malanche	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	1420.45

Name	Base Legal	Recognition Date	Extension (Ha)
Caringa	RDE N°153-2018-MINAGRI SERFOR-DE	18/07/2018	2178.61
number	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	176.06
Cicasos	RDEN°153-2018-MINAG RI SERFOR-DE	18/07/2018	1380.82
Gaudy	RDE N° 153-2018-MINAGRI SERFOR-DE	18/07/2018	122.8
Total			19 435.8

(*) The Villa María hill, which is part of the ACR, is part of the area that includes the Villa María del Triunfo hill included in the sectoral list of fragile ecosystems of the MINAGRI.

(**) The Amancaes hill, which is part of the ACR, is part of the Amancaes hill included in the sectoral list of fragile ecosystems of the MINAGRI.

(***) Lomas Carabayllo I and II, which are part of the ACR, are located within the Carabayllo hill included in the sectoral list of fragile ecosystems of the MINAGRI.

Table 3 Sectoral list of fragile ecosystems from MINAGRI Coastal hills in the Province of Lima
Source: SERFOR

The hills included in the MIDAGRI sectoral list of fragile ecosystems (Table 3) also have conservation instruments.[31]. Likewise, within the framework of the measures adopted for the conservation of the coastal hills in the Province of Lima, to date the district municipalities of Independencia, Carabayllo, Villa María del Triunfo, Pachacámac and Punta Hermosa, have ordinances that declare district interest in the conservation of the hills that are within its jurisdiction.

1.6.4.2. Wetlands

Coastal wetlands are defined as extensions or surfaces covered or saturated with water, under a natural or artificial water regime, permanent or temporary, fresh, brackish or salty,

and that house characteristic biological communities and are found along the coastal coastline. and maritime[32]. In this sense, mangroves, lagoons, estuaries, lagoons, deltas, oases and swamps are considered coastal wetlands (MINAM, 2019).

Wetlands are also considered fragile ecosystems.[33], and the State recognizes its importance as a habitat for species of flora and fauna, particularly migratory birds, prioritizing its conservation. This ecosystem is the only one that has been the subject of an international treaty: "Convention on Wetlands of International Importance", known as the Ramsar Convention, which in the case of the province of Lima acquires importance for the protection of the habitat of waterfowl.

[31]Executive Directorate Resolution No. 253-2018-MINAGRI-SERFOR-DE, which approves the "Conditions for the use of forest and wildlife resources in the ecosystems included in the sectoral list of fragile ecosystems" D.S. N°007-2020-MINAGRI, which approves the Interinstitutional Action Protocol to Manage and Protect fragile ecosystems.

[32] Supreme Decree No. 004-2015-MINAM, which approves the National Wetlands Strategy

[33] As described in sections 99.2 and 99.3, of article 99 of Law No. 28611, General Environmental Law.

1.6.4.3. Pantanos de Villa Wildlife Refuge

To date, the Province of Lima has the Los Pantanos de Villa Wildlife Refuge, categorized as a Protected Natural Area since September 1, 2006, through Supreme Decree N°055-2006-AG. It has an area of 263.27 ha and aims to conserve a representative sample of the Subtropical Pacific Desert Swamps, including important plant communities representative of coastal wetlands, as well as migratory and resident birdlife.

Since 1997, the Pantanos de Villa Wetland has been recognized as "Wetland of International Importance or Ramsar Site No. 884" by the Ramsar Convention, for being a feeding, reproduction, lodging, resting and refuge area for resident and migratory Nearctic birds. Many of these migratory birds settle during the spring and summer of the southern hemisphere, distributing themselves along the coasts, high Andean lagoons and Amazonia of the region.

The Los Pantanos de Villa Wildlife Refuge has a Master Plan which directs its objectives to the conservation of its biological diversity and to promote inter-institutional cooperation for the benefit of Los Pantanos de Villa. Said Plan contains the zoning and conditions of use within the ANP, establishes strategies in the environmental, economic and social aspects, the control and surveillance system, environmental monitoring, recovery of degraded areas, the physical demarcation of the ANP, among others. It is important to note that Los Pantanos de Villa has a headquarters of the National Service of Natural Areas Protected by the State –SERNANP, made up of a Chief, specialists and park rangers. Likewise, within the ANP there are 4 owners, which were there before it was declared ANP, with the Lima Parks Service – SERPAR, the one that manages the largest extension of hectares, with almost 80% of the total. Currently, the Municipal Authority of the Pantanos de Villa - PROHVILLA, by agreement with SERPAR, is the one who manages this space called the Los Pantanos de Villa Metropolitan Ecological Area.[34], in which it continues to carry out the activities of (1) Environmental education, (2) scientific

research and (3) ecotourism as a means for the conservation of the area.

PROHVILLA is a decentralized public body of the Metropolitan Municipality of Lima, with legal, administrative and economic autonomy, whose mission in accordance with its functions and powers established in its Statute, approved by Ordinance No. 838 (September 22, 2005), is "Work for the conservation of the Pantanos de Villa wetland – Wildlife Refuge, through adequate technical, legal and administrative management in the Pantanos de Villa Special Regulation Area – ZRE PV, promoting the active participation of all partner actors economic resources present in the territorial environment of the wetland, in order to sustainably protect the ecosystem and its biodiversity.

Along these lines, in 2020, Ordinance No. 2264-MML, "Special Regulation Zone of the Villa Reservoirs (ZRE PV)", was promulgated with the objective of promoting the conservation and sustainable development of the territorial environment of the Wetland. Pantanos de Villa, which has an area of 2,572 hectares and is divided into (08) units of

[34] Metropolitan Council Agreement No. 26-1991.

environmental planning, in order to organize and harmonize the occupation and use of space (see Figure 15).

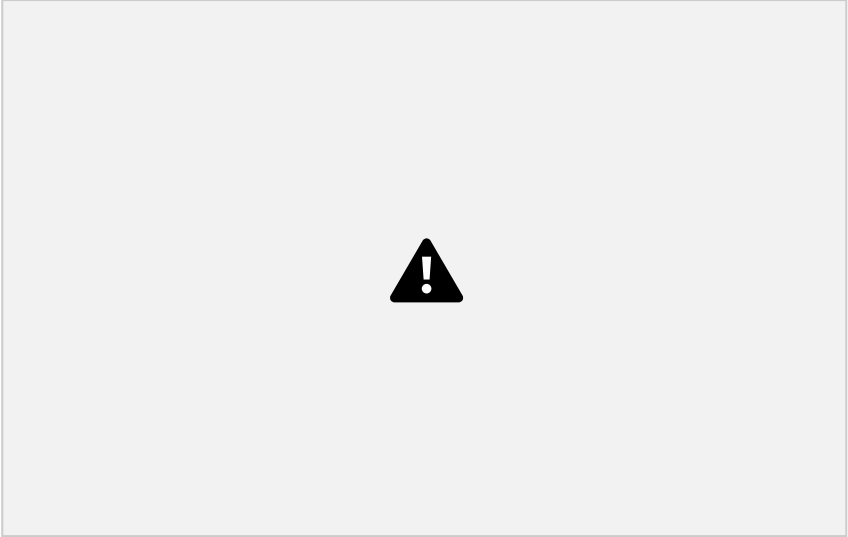

Figure 15. Environmental Management Units of the ZRE PV.
Source: Ordinance No. 2264-MML.

The Board of Directors of PROHVILLA approved as management instruments for the ZRE PV the plans for: (1) Communication, education, awareness and participation of the Municipal Authority of Los Pantanos de Villa 2020-2024, (2) Control and surveillance plan in the ZRE PV, (3) ZREPV water infrastructure recovery plan, which includes 12 canals, 3 puquials and 1 relict wetland of approximately 7 ha.

Regarding its biodiversity, PROHVILLA has historical records of monitoring carried out in Los Pantanos de Villa from 2011 to the present, where 211 have been identified[35] bird species

among residents[36] and migratory[37]. Regarding mammals, 8 species have been recorded, 3 flying[38] and 5 land[39]. In reptiles, 6 species have been recorded[40]; while in the case of fish, 14 species have been identified. On the other hand, according to the monitoring carried out by PROHVILLA since 2019, 65 species of vascular plants have been reported, including native, introduced, naturalized introduced and potential invasive, among the most common species of vascular flora in Villa swamps we have " Wild tomato", "Totora balsa", "Water lettuce", "Totora enea" and "Elephant ear". This plant cover feeds on nutrients from the subsoil, rich in salts and organic matter accumulated over many years.

Likewise, PROHVILLA, in its technical work called "Towards management with an ecosystem approach of the Pantanos de Villa wetland: Formulation of the Diagnostic Analysis of the Pantanos de Villa Wetland 2019", spatially identified each of the threats towards

35 Pulido (2018): "One hundred and fifteen years of bird records in Pantanos de Villa"

36 Most notable resident species: "Red Duck", "Meergant" and "Seven Colors".

37 Most notable migratory species: "Franklin's Gull", "Yellow-legged Least Sandpiper", "Spiking Sandpiper", Gray Gull", "Andean Gull", "Common Parihuana".

38 Flying mammals: Artibeus fraterculus, Tadarida brasiliensis and Myotis Atacamensis

39 Mamíferos terrestres: Akodon sp., Mus musculus, Rattus rattus, Rattus norvegicus y Cavia tschudii"

40 Reptiles: Stenocercus modestus, Microlophus thoracicus, Microlophus peruvianus, Pseudalsophis elegans, Phyllodactylus microphyllus y Rhinella marina

the wetland, considering its impact on biodiversity and the ecosystem, the most notable being: impact on the quality of the water resource, inadequate management of solid waste, inadequate urban activities, among others.

It is important to note the existence of a relict wetland located in the La Chira sector, a vestige of what was at the beginning of the 20th century the extension of the Pantanos de Villa wetland (according to Pulido & Bermúdez, Pantanos de Villa covered an area of 2000 hectares[41]). Likewise, it is necessary to highlight that within Metropolitan Lima, in addition to Pantanos de Villa, there are other coastal wetlands, which, although they do not form protected natural areas, are important for all the benefits they represent for the city. The list of these wetlands, including Pantanos de Villa, is shown in Table 4 and Figure 16.

N°	Wetlands	UTM coordinates*			Level of Protection
		Zone	It is	North	
01	Punta Wetland black	18L	306242.95	8633312.58	Without protection
02	Quilcay Wetland	18L	295059.78	8641245.55	Without protection
03	Mamacona Wetland	18L	292262.56	8644228.56	Without protection
04	Pantanos de Villa Wetland	18L	283555.08	8649039.79	ANP Wildlife Refuge**
05	Left of the humidal La Chira	18L	279268.97	8650129.34	Without protection
06	Santa Rosa Cemetery Lagoon	18L	284139.98	8652515.28	Without protection

07	Molina Lagoon	18L	290981.45	8663532.63	Without protection
08	Ancón Wetland	18L	263108.96	8697424.00	Without protection

Table 4 Location of wetlands in Metropolitan Lima.
Source: NGO Cooperación.

* Date WGS94
** Declared under Supreme Decree N°055-2006-AG

[41] Pulido, V & Bermúdez, L. 2018. Current status of the conservation of the habitats of the Pantanos de Villa, Lima, Peru. Arnaldoa 25(2): 679-702.

Figure 16. Map of the wetlands of Metropolitan Lima based on a registry carried out by the NGO CooperAccion.
Source: NGO Cooperación.

1.6.4.4. coastal desert

It is an arid to hyperarid ecosystem with areas largely devoid of vegetation, made up of sandy soils or with rocky outcrops, with notable plant formations such as tillandsiales (rosettes), cacti area (columnar, prostrate and globular), bushes, among others. (MINAM, 2019). They occupy significant areas in the Province of Lima, and given their physiography and extension, coastal deserts can host and give rise to other ecosystems.

Currently, the coastal desert is highly threatened, mainly due to disorganized population growth. Among other problems are agricultural, industrial, mining activity, land ownership and change in land use. These actions result in the fragmentation of desert habitat, threatening the health of endemic species, such as geckos. Therefore, it is important to conserve the areas that record the biodiversity, flora and fauna, typical of this ecosystem.

Being a coastal city, the coastal marine area in Lima is of great importance from a social, economic and natural point of view. The ocean provides socio-economic benefits

economic through productive activities such as fishing, transportation and the marketing of

goods. In addition, it is a primary source of food for the citizens of Lima, and of great importance for food security. Likewise, they provide ecosystem services such as the production of oxygen and the absorption and redistribution of natural and anthropogenic carbon dioxide (MINAM, s.a.). Climate change has important consequences on coastal marine areas. Increases in the average temperature of the oceans have been observed since the 1970s by absorbing 90% of the excess heat of the climate system, their acidification due to the concentration of CO2 and the loss of oxygen. Additionally, the loss of ice has increased the average sea level worldwide (IPCC, 2019). The rise in sea level is one of the dangers facing the Province of Lima.

1.6.4.5. Andean scrub

Ecosystem with woody shrub vegetation of variable composition and structure (including formations of cacti or cardonales) with a height of no more than 4 meters and whose soil coverage is greater than 10%, in the same way they present trees distributed in a dispersed manner (MINAM, 2019).

This ecosystem constitutes the habitat of numerous species, preserving its biodiversity; On the other hand, it has a special relevance for being a source of fuel for local populations (due to the use of woody species for the production of wood and firewood). However, among its main threats are the loss of habitat due to the development of mining and agricultural activities, overgrazing and the excessive exploitation of shrubs and other plants for use as fuel.

1.6.4.6. Rivers

It is a natural current of water of variable depth and size that normally flows with continuity; It can be located on flat or gently sloping reliefs up to extremely rugged reliefs and high slopes. It has a flow that is rarely It is constant or regular throughout the year, and can reach minimum levels in the season drought (MINAM, 2019).

Regarding its biodiversity, as part of the flora are the communities of columnar cacti, pluvifoliate shrubs, stony rocks with mixed vegetation, grassland and shrub steppe. For its part, the fauna is represented by birds, among them the “Band-tailed seedeater”, “Lesser goat clipper”, “Spotted dove”, among others. In As for mammals, they are represented by “Muca muca”, “Vizcacha”, among others. By province of Lima three main rivers cross, Chillón, Rímac and Lurín. About them described in greater detail in the water resources item.

1.6.4.7. Green areas and urban trees

Green areas are made up of public or private areas or spaces enabled with vegetation, those spaces reserved for green areas in accordance with the Metropolitan Urban Development Plan and those natural spaces, which define the

ecosystems in the jurisdiction of the Province of Lima. They are comprised of cover plants, shrubs, flower beds, palm trees, trees, among others.

Green areas for public use are of an intangible, inalienable and imprescriptible nature, so their conservation, protection, and maintenance are actions that, for reasons of public health, quality of life, collective well-being, ecological balance and urban sustainability, are part of the Metropolitan Environmental Policy. Therefore, the Metropolitan Municipality of Lima and the district municipalities are obliged to conserve, defend, protect and maintain in good condition the green areas for public use under their jurisdiction.

Green areas and urban trees provide multiple environmental services to citizens, such as the generation of oxygen, reduction of pollutant levels in the air, noise dampening, climate regulation, reduction of soil erosion, in addition to representing sites shelter, protection and feeding of wildlife. Likewise, they provide social services by constituting spaces for coexistence, recreation, and active and passive recreation for their inhabitants, contributing to the improvement in the quality of life of their inhabitants.

The province of Lima currently has an authorized green area surface of 29,822,768.89 m^2, and a per capita index of 3.5 m^2 of green area per inhabitant approximately. This green area surface is made up of district green areas and those that are under the administration of the Municipal Companies and Decentralized Public Organizations of the MML, such as SERPAR-LIMA, PROHVILLA, APCV, PATPAL FBB, EMILIMA S.A., EMAPE S.A. and the ATU.

Because the province of Lima is located on a coastal desert, green areas and urban trees require watering for their permanence, since precipitation is low and therefore normally not enough to keep the vegetation alive. The different municipalities are responsible for watering the green areas under their jurisdiction.

The province of Lima presents a deficit of 47,029,694.3 m^2 of green area, based on what is recommended by the World Health Organization-WHO, which indicates that cities have a minimum of 9 m^2 Per inhabitant[42]. Likewise, the presence of 1 tree for each inhabitant is recommended.[43] of the city, which generates a demand for 25,724,922 trees in Metropolitan Lima.

Green areas and urban trees are of vital importance to address climate change in cities. Natural spaces, especially trees, absorb carbon dioxide from the atmosphere, being a carbon sink. Likewise, it allows reducing temperatures in urban areas to deal with heat waves by generating shadows and absorbing less heat compared to the asphalt. In the case of green areas located near
rivers, allow water to infiltrate more quickly, reducing the possibility of flooding and increasing the recharge of the water table. Green areas also have benefits on the health of the population by improving air quality (Robbel, s.a.).

[42] Sorensen, Mark, Valerie Barzetti, Kari Keipi and John Williams (1998), Urban Green Area Management, Good Practice Paper 109, IDB, Washington, United States of America.

[43] Gámez, Vicente (2005), "On systems, typologies and standards of green areas in urban planning", Journal of Urban and Landscape Design, (2) 6, Universidad Central de Chile, Santiago, Chile, pp. 1-22, http://dup.ucentral.cl/pdf/000002.pdf

1.6.4.7.1. Lima Parks Service – SERPAR

The Lima Parks Service – SERPAR, is a decentralized public organization of the Metropolitan Municipality of Lima.[44], its function is the promotion, organization, administration, development and maintenance of metropolitan parks, zonal clubs, zoos and botanicals of the Province of Lima, for recreational, cultural, sports and environmental preservation purposes. SERPAR currently manages 10 zonal clubs and 9 metropolitan parks, whose green area is 191.40 ha. It also has three nurseries located in the zonal clubs Huáscar (Villa El Salvador), Sinchi Roca (Comas) and Cahuide (Ate) where all the flowers, plants and trees that it distributes throughout the city.

In the green areas and urban trees in the zonal clubs, metropolitan parks and the reserve park, a total of 79 species of birds have been recorded to date, with the points with the greatest species richness being the Ancón Forest with 65 species. , Huáscar zone club with 41 species, Exhibition Park with 29 species, Huayna Cápac zone club with 27 species, Reserve Park with 27 species and the Rings Park with 22 species.

1.6.4.7.2. Conclusion

Ecosystems and green areas are crucial spaces in urban areas, since they have many benefits for the populations that live there. They provide ecosystem services with various economic, social and environmental benefits. They highlight the health benefits of reducing respiratory diseases, the possibility of having recreational spaces for the population and its importance for the city's climate management.

The province of Lima has the particularity of being located on a coastal desert with diverse particular ecosystems, including coastal hills, wetlands, rivers and the marine-coastal zone. The hills have been characterized as fragile ecosystems, so their protection is of great importance. These physical characteristics also imply the need for resources to irrigate green areas and urban trees as part of their maintenance.

Ecosystems and green areas have an impact on both mitigation and adaptation to climate change. They help reduce GHG emissions by capturing CO_2; while they contribute to the reduction of temperature and water infiltration to deal with heat waves and floods respectively.

1.6.5. Economical

The economy of Metropolitan Lima contributes about a third of the gross domestic product, concentrating about 80% of the country's economic activities and financial services. This section allows us to identify the sectors of the city where economic activities are concentrated, which is pertinent given the spatial stratification of the social gap identified for the city.

[44] SERPAR was created by Mayor's Decree No. 031 of 1984.

Considering the 2017 Census, it can be indicated that the Working Age Population (PET; population between 15 and 59 years old) represents just over 69% of the total inhabitants of the province. Within the prospective population analysis of Metropolitan Lima, data on the economic development capacity of the municipalities are considered through the Economically Active Population (EAP).

Until before the COVID-19 pandemic, the unemployment rate remained stable at around 6% over recent years in Metropolitan Lima, as shown in the following Figure 17.

Figure 17. Evolution of the unemployment rate, according to moving quarters 2014 – 2020. Source: (INEI, 2020)

This unemployment indicator is related to the degree of preparation of the population to participate actively and qualified in the activities that may be planned to promote the economic development of the territory, assuming that the higher the degree of qualification, the greater the possibility of economic development. in the city. The percentage of PEA registers 48% (INEI 2017) (IDOM, 2020).

According to the IV National Economic Census-CENEC (INEI 2008), Lima Centro concentrates the majority of employment in all sectors (82% of activity in the primary sector, 61% in the secondary sector and 66% in the tertiary sector), such as is shown in Table 5.

Interdistrict Area	Primary sector	Secondary sector	Third sector
Lima Center	82%	61%	66%
Five	2%	24%	13%
Five	4%	9%	12%
Five	12%	6%	10%
Total	100%	100%	100%

Table 5 Employment in economic sectors by interdistrict area. Source: (INEI, 2008).

Taking this information into account, we can see how the pattern of distribution of establishments and employment in the Metropolitan Area is especially concentrated in the Lima Center area, specifically in the Cercado de Lima district, which has the largest number of establishments (50,997) located for the most part in the Historic Center.

A pattern of distribution of economic activities linked to the main structuring axes of the city is also observed, as is the case of Avenida Argentina and Avenida Arequipa, in Lima Centro, Panamericana Norte and Avenida Túpac Amaru in Lima Norte and the Huaylas Avenue in South Lima.

The Lima Sur sector is presented as a still underdeveloped area, whose highest concentrations are located in the Villa El Salvador industrial park (IMP 2014). In addition to the aforementioned sectors, agricultural activity must be considered, which, despite constituting a small part of the Economically Active Population, has an important relevance in the Metropolitan Area.

Figure 18. Concentration of establishments and jobs.
Source: (IMP, 2014).

Regarding agricultural activity, it is located in the North, East and South areas, associated with the valleys of the Chillón, Rímac and Lurín rivers, respectively, quite far from the Central districts. Although this economic activity constitutes a small part of the Economically Active Population, it has an important relevance in the Metropolitan Area.

Figure 19. Agricultural valleys.
Source: (IMP, 2014).

A relevant aspect of the economy of the metropolitan area is that the spatial stratification of the social gap has been identified, through the GINI coefficient.[45], since it is observed that:

- The highest inequality indices are found towards the northern, southern and eastern ends of these interdistricts.
- Inequality is characterized by having the lowest levels in peripheral districts, with lower levels of density and greater rurality.
- Some of the most central districts such as La Victoria, Cercado de Lima and Chorrillos have low levels for the GINI coefficient.
- The districts with the lowest levels of inequality are found in the districts of Central Lima, with coefficients between 0.26 and 0.27.

At the level of economic income, spatially it is also identified that Central Lima concentrates the population with a higher level of income, and in Eastern Lima, this indicator is only available for the district of La Molina; while, towards the periphery of the city, the level of per capita income is reducing from medium to medium-low, in the areas of North, South and East Lima.

1.6.6. Social

The province of Lima has 8,574,974 inhabitants, representing 27.45% of the country's

population and 90.40% of the population of the department of Lima, according to data from the 2017 National Census. The districts that concentrate the largest number of inhabitants are San Juan de Lurigancho with 1,038,495, San Martín de Porres with 654,083, Ate with

[45] The GINI Coefficient is a measure to calculate inequality, usually used to estimate income inequality.

599,196 and Comas with 520,450 inhabitants respectively. The district with the smallest population is Santa María del Mar with 999 inhabitants.

Of the total population registered in Metropolitan Lima, 48.6% are men, while 51.4% are women. For the intercensal period 2007-2017, it can be seen that the male population increased by 12.3% in ten years and the female population by 13.2% during the same period. By age groups, between 0 and 19 years there are more men than women, with the group from 1 to 4 years of age being the one with the highest masculinity index, that is, 104 men for every 100 women. In the other age groups there are more women than men (see Figure 20).

Figure 20. Population pyramid.
Source: (INEI, 2017).

The demographic density in Lima is concentrated in the district of Breña with a population density of 26,526.43, followed by Lince 19,894.91, Surquillo 19,680.65, La Victoria 19,038.38 and Santa Anita with 18,375.54 inhabitants per square kilometer, according to information from the 2017 Census. According to the ninth Report on Indicators on Quality of Life carried out by Lima Como Vamos, based on the figures collected from the population identified with DNI, it is estimated that the districts that host the largest number of migrants are San Juan de Lurigancho , San Martin de Porres and Ate. In total there are approximately 2,814,757 migrants residing in Metropolitan Lima.[46]

[46] It should be noted that these figures are not exact, given the presence of non-naturalized foreign citizens and Peruvian citizens who have not updated their place of residence, but live in one of the districts of Metropolitan Lima or Callao.

On the urban area, unplanned urban expansion has resulted in an extensive city (more than 70,000 hectares of surface in 2014) and low associated density (119 inhabitants/ha in 2014). The Lima Centro area filled up in the 1990s, while the most recent growth has taken place in the outer areas of the city, in north, east and south directions. In this expansion process, informal urban development has predominated, in some cases occupying the banks of rivers and areas of high slopes.

Figure 21. Evolution of the urban footprint 1954-2014.
Source: (IMP, 2014).

1.6.6.1. Education

For the analysis of educational facilities in Metropolitan Lima, the registry of educational institutions available in the download center of the Ministry of Education (MINEDU, 2020) has been taken. Taking this information, it is obtained that there are a total of 17,287 training centers in the Metropolitan Area. The breakdown of the number of centers by educational level and by area is provided in Table 6.

Area interdistrital	Home	Primary	Secondary	Basic alternative and special	Technique productive
Lima Center	1599	803	591	129	234

East Lima	2436	1366	814	25	177
North Lima	2430	1553	892	32	188
Lima Sur	1879	1009	563	28	163
Total	**8344**	**4731**	**2860**	**214**	**762**

Table 6 Training centers by educational level and interdistrict area.

Source: (MINEDU, 2020).

As occurs with the other facilities and infrastructure analyzed, it can be seen that the density of educational facilities decreases towards the periphery of the city (see Figure 22). Furthermore, it is observed how higher education educational centers are concentrated in central Lima.

Figure 22. Density of educational centers.
Source: (MINEDU,2020)

1.6.6.2. Health

In the registry of health facilities of the Ministry of Health (MINSA) for the year 2019, it is obtained that there are a total of 2,424 health facilities, distributed in 165 health establishments with confinement, 1,905 health establishments without confinement and 354 Support medical services, according to the categories established by MINSA. The distribution by interdistrict area is shown in Table 7.

Interdistrict area	Establishments of health with	Establishments of health without	medical service support

	internment	hospitalization	
Lima Center	92	654	230
East Lima	23	526	56
North Lima	24	362	43
Lima Sur	26	363	25
Total	**165**	**1905**	**354**

Table 7 Health facilities by category and interdistrict zone.
Source: (MINSA, 2020)

1.6.7. Cultural

Lima's cultural heritage is the result of the social, economic and historical processes that the city has experienced since its pre-Hispanic occupation until its current configuration as the capital of the republic. As an example of the great cultural heritage it houses, the Historic Center of Lima (CHL) was declared a World Cultural Heritage by UNESCO in 1991.

The Peruvian state, having signed the “Convention on the Protection of the World Cultural and Natural Heritage”, approved in 1972 by UNESCO and ratified by Legislative Resolution No. 23349, is obliged to protect cultural heritage from all its state agencies. including the Metropolitan Municipality of Lima.

Although the CHL is one of the most important assets for the city, where we find the Casona San Marcos or the University Park; the Plazas Mayor, San Martín, Italy; the Fifth Dam; the Court of the Holy Inquisition; museums; the Government Palace, the Municipal Palace, the Torre Tagles Palace; the Cathedral of Lima, as well as churches and convents. The historic centers of the districts of Pueblo Libre, Barranco and Comas are also part of this cultural legacy.

The city has the Master Plan for the Historic Center of Lima 2019-2029 (with a vision to 2035). It was prepared by PROLIMA and promotes the historical, architectural, urban, social and human recovery of the CHL. It is a specific plan that establishes the guidelines for any intervention that is to be carried out in the Historic Center during a period of 10 years: land uses, public or private projects, inspection, etc. with the aim of having a single and comprehensive work horizon.

In Metropolitan Lima, 334 archaeological zones from different historical periods have been identified, which places the city in a privileged place of cultural wealth. It brings together time periods of regional lordships, the formative period, pre-Hispanic occupations and the late period. The districts with the greatest number of archaeological zones are: Pachacamac with 24, Miraflores 22, San Isidro 37 and San Juan de Lurigancho with 47 archaeological zones respectively.

The most notable archaeological zones in the city include:

- Garagay (San Martin de Porres)
- Huallamarca (San Isidro)
- Huantille (Cieneguilla)
- Mateo Salado (Lima)
- Pucllana (Miraflores)
- Puruchuco (Ate)
- San Borja (San Borja)
- St. John of Pariachi (Ate)
- Santa Catalina (La Victoria)

For the adequate protection of cultural heritage, the constant urban growth of the city, the use of these areas, formal urban development and ensuring protection measures for archaeological areas and their surroundings must be adequately managed.